Applied Soft Computing Journal 190 (2026) 114585

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Natural language processing for construction engineering: Artificial intelligence-driven text analysis techniques

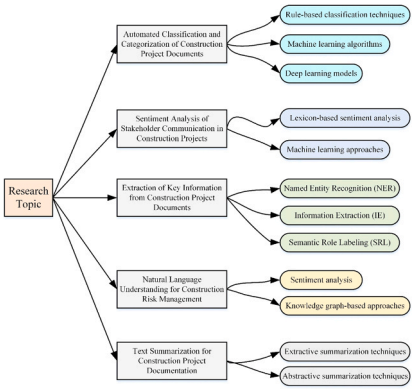
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HIGHLIGHTS

- AI-driven NLP enables scalable analysis of unstructured construction engineering texts.
- Transformers, RLHF, and XAI advance reliability of NLP for safety and risk tasks.
- Systematic review maps NLP methods, datasets, and applications from 2001–2025.
- Identifies gaps in benchmarks, datasets, and deployment of LLMs in construction.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Artificial intelligence-based text analysis
Natural Language Processing
Potential implications
Automated document classification
Sentiment analysis
Explainable artificial intelligence

ABSTRACT

The construction engineering sector has witnessed significant advancements through the integration of Natural Language Processing (NLP), driven by artificial intelligence-based text analysis methods. As construction projects generate vast volumes of unstructured textual data ranging from contracts and specifications to stakeholder communications extracting actionable insights from this data has emerged as a critical challenge. This study presents a comprehensive review of scholarly literature from 2001 to 2025, systematically analyzing trends, methodologies, and applications of NLP in construction engineering. The findings reveal a sharp increase in research activity, with growing emphasis on advanced AI techniques such as deep learning models, transformers, multimodal analysis, and explainable AI. These approaches have enabled enhanced document management, improved risk assessment, real-time decision-making, and more effective stakeholder engagement. Moreover, the study identifies key challenges, including limited access to domain-specific datasets, model interpretability, data interoperability, and the scalability of NLP applications. Furthermore, the study highlights the emerging need to evaluate large language models (LLMs) using domain-specific benchmarks to ensure their reliability and practical applicability in construction NLP tasks. By addressing these limitations and highlighting future research directions, this review underscores the transformative potential of NLP in reshaping construction engineering practices and supports the development of intelligent, data-driven solutions tailored to the unique complexities of the construction domain.

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https://doi.org/10.1016/j.asoc.2026.114585
Received 1 May 2025; Received in revised form 13 November 2025; Accepted 23 December 2025
Available online 8 January 2026
1568-4946/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

The construction industry, playing a pivotal role in the global economy, actively contributes to infrastructure development and shapes the built environment [1]. However, this industry grapples with numerous challenges, including the sheer volume of information, complex documentation, and the need to extract valuable insights from extensive unstructured textual data [2,3]. To address these challenges, the construction engineering field has witnessed the emergence of Natural Language Processing (NLP) as a prominent domain of study and advancement. The rapid progress and widespread adoption of AI and its applications have led to an increased use of AI-driven text analysis techniques in construction [4,5]. These techniques aim to enhance project management, decision-making, and overall efficiency by providing more efficient and proficient methods for analyzing and understanding textual data.

Although the construction sector contributes about 13 % to global GDP, it remains among the least digitized industries, leading to underutilization of valuable data generated from project documentation, stakeholder communication, and operational processes [6]. In addition, project documentation generally encompasses extensive quantities of text-based information, encompassing project plans, specifications, contracts, and records of communication [7]. The process of extracting valuable insights from textual sources is a demanding and time-consuming endeavor that frequently necessitates substantial human labor and expertise which increases the costs by more than 30 % in some cases [8,9].

Within the field of construction engineering, the effective handling of extensive and varied textual data has become a prominent and urgent issue. Construction projects need a substantial volume of paperwork, including a wide array of materials such as project specifications, contractual agreements, reports, and contact with stakeholders. The extraction of relevant insights and information from unstructured textual data is of utmost importance in order to facilitate informed decision-making, enhance project management efficiency, and achieve better results. The emergence of NLP has provided a revolutionary answer to this issue, enabling experts in the construction industry to use the capabilities of text analysis. In order to tackle these challenge [10], scholars and professionals have progressively resorted to employing NLP methodologies, driven by the progress made in AI, machine learning, and deep learning [11]. NLP encompasses the utilization of computational techniques to comprehend, decipher, and produce human language [12,13]. The utilization of AI-powered text analysis techniques has facilitated the automation and optimization of various tasks within the construction industry, including document classification, information extraction, sentiment analysis, and summarization, among other functions [14].

There is a gap within the existing body of knowledge pertaining to the utilization of NLP in the field of construction engineering [15,16]. Despite emerging interest, a substantial research gap persists in the integration of NLP in construction engineering. Recent studies, including the review by Hussain et al. (2024), have pointed out that much of the existing literature remains fragmented, with limited use of advanced AI methods and inadequate focus on deployment in real-world AEC scenarios. Moreover, their analysis emphasizes the need for deeper exploration of transformer-based architectures, multimodal analysis, and domain-specific model training [17]. These findings reinforce the rationale for this present study, which aims to deliver an updated, targeted, and AI-integrated review covering both scholarly evolution and practical trends in the field. Although NLP techniques have made significant progress in various domains such as healthcare, finance, and customer service, their implementation in the field of construction engineering is still in its early stages [18]. The current body of scholarly work primarily emphasizes conventional approaches to data analysis, such as statistical analysis and data mining, while the potential of NLP techniques remains largely unexplored [19,20]. Hence, the purpose of this review is to address the existing research void by examining the recent progressions in AI-based text analysis methods that are specifically applicable to the field of construction engineering [21–23].

There is also an ongoing paradigm shift in the discipline of Construction Engineering, a vital discipline as it impacts global development of infrastructure and overall worldwide economic growth. The emergence of NLP and all of its uses have shown great potential to transform how we practice in Construction Engineering. Unfortunately, there are several limitations within the existing body of scholarly work related to NLP in the field of Construction Engineering. Thus, this study focuses on addressing some of those limitations. Previous literature has addressed the topic of NLP in the construction industry through review articles, but they have their own set of limitations that require a more comprehensive and contemporary analysis.

- **Incompleteness:** The issue of incompleteness arises when examining the current review articles in this field, as they provide significant insights but often lack to give a thorough overview of the most recent breakthroughs in AI-driven text analysis approaches. The dynamic and swiftly progressing field of NLP necessitates a comprehensive examination that incorporates current advancements.
- **Lack of Specificity:** The absence of specificity is evident in several previous assessments, as they provide broad observations about the use of NLP in various sectors, without specifically addressing its application within the field of construction engineering.
- **Future Research Directions:** Future research directions are often neglected in previous evaluations, since they lack to emphasize the need of identifying topics for further investigation and developing trends.

This study aims to provide a comprehensive analysis of NLP in the context of the construction engineering domain, with a focus on summarizing existing advancements and exploring future trends and potential applications. The study has mainly two objectives, (1) To review and analyze the current state of NLP applications in the Architecture, Engineering, and Construction (AEC) industry, summarizing the key advancements and trends and (2) To explore potential future trends in NLP within the AEC industry, with a particular emphasis on emerging applications and their implications.

This paper addresses the aforementioned gap by providing a thorough examination of AI driven text analysis methods within the domain of construction engineering. Additionally, the emphasis on future research paths adds to its value as a useful resource for researchers and industry practitioners alike. This work makes a valuable contribution to the advancement of NLP in the construction industry by addressing the existing constraints and offering a more specialised, up-to-date, and methodologically rigorous assessment. This study makes a distinctive addition by conducting a comprehensive examination of the most recent breakthroughs, possible ramifications, and recommendations for future research in order to ensure that NLP effectively addresses the changing requirements of the construction industry.

The emergence of these trends in NLP research indicates a large-scale shift toward the adoption and use of information technology and digitalization in construction engineering. The obvious positive effects of NLP have been demonstrated in the area of construction engineering; therefore, it is essential that a complete and detailed review of the advancements that have taken place be conducted. Therefore, this investigation fills an existing knowledge gap by providing a thorough analysis of the advancements that have occurred in AI-based text analysis methods with an emphasis on the latest developments in construction engineering. In addition, this investigation identifies the potential ramifications of using NLP as a means to address the many problems currently being experienced by the industry. NLP can lead to the transformation of construction engineering practices through the augmentation of decision-making, improvements in the efficiency of

managing projects, and the extraction of valuable insight from unstructured textual data. Therefore, this investigation provides a systematic and focused review of current trends, issues, and opportunities related to the use of NLP to enhance understanding of how it can transform the building industry.

2. Methodology

2.1. Data collection

Within this review duration, there has been a growing body of academic inquiry into the application of NLP technology within the discipline of Construction Engineering. Fig. 1 illustrates the methodological steps that were undertaken. The methodology employed in this research included an exhaustive and thorough review of all literature references. A total number of publications within the time period of 2001–2025 was identified through a keyword search relevant to NLP applications in construction engineering. An extensive and broad-based examination was conducted across reputable academic journals such as IEEE Xplore, Springer, ScienceDirect, Scopus, ACM Digital Library, and Google Scholar to ensure a complete selection of applicable materials.

For literature search the study utilized a keyword string consisting of the following key terms: "Natural Language Processing," "NLP," "Construction Engineering," "Construction Industry," "Text Analysis," "AI-driven Analysis," "NLP Applications," "Deep Learning Models," "Transformers," "Document Management," "Risk Assessment," "Stakeholder Interaction," "Decision-making Processes," "Cooperation," "Multimodal Analysis," "Explainable AI," "Domain-specific Models," "Real-time Applications," "Building Information Modelling," "BIM," "Construction NLP," "Construction Text Mining," "Sentiment Analysis," "Construction Document Classification," and "Construction Natural Language Understanding."

The inclusion and exclusion criteria developed to guide the selection of studies reviewed in this study ensured both relevance and rigorous methodology for each of the selected studies. The inclusion criteria indicated that studies would be considered eligible for inclusion in this review if they utilized either NLP or Artificial Intelligence (AI) – driven methods of analyzing text; were written in an English language peer-reviewed journal; and had been written between the years of 2001–2025. Articles were excluded during the initial screening phase based upon their lack of focus on the use of NLP, lack of relationship to Construction Engineering or the Built Environment, and/or insufficient information related to the methodologies used. Additionally, articles that were not scholarly (i.e., editorials, commentaries, short notes, etc.) and/or presented opinions without empirical evidence were also excluded. This process resulted in a total of 177 articles representing high quality domain relevant research being included in the final review.

All studies included in the systematic literature review underwent a clearly defined and objective screening protocol to maximize the quality and relevance of the studies included in the review as shown in Fig. 2. The Identification stage involved conducting searches on major academic databases using pre-defined search terms which resulted in the retrieval of 738 study records. Following the removal of duplicate records, there were 512 records remaining for evaluation in subsequent stages.

The Screening stage entailed an examination of the title and/or abstract of each of the 512 records. All records whose titles and/or abstracts indicated a lack of alignment with the review's purpose (i.e., a study that is irrelevant to NLP or does not relate to the construction

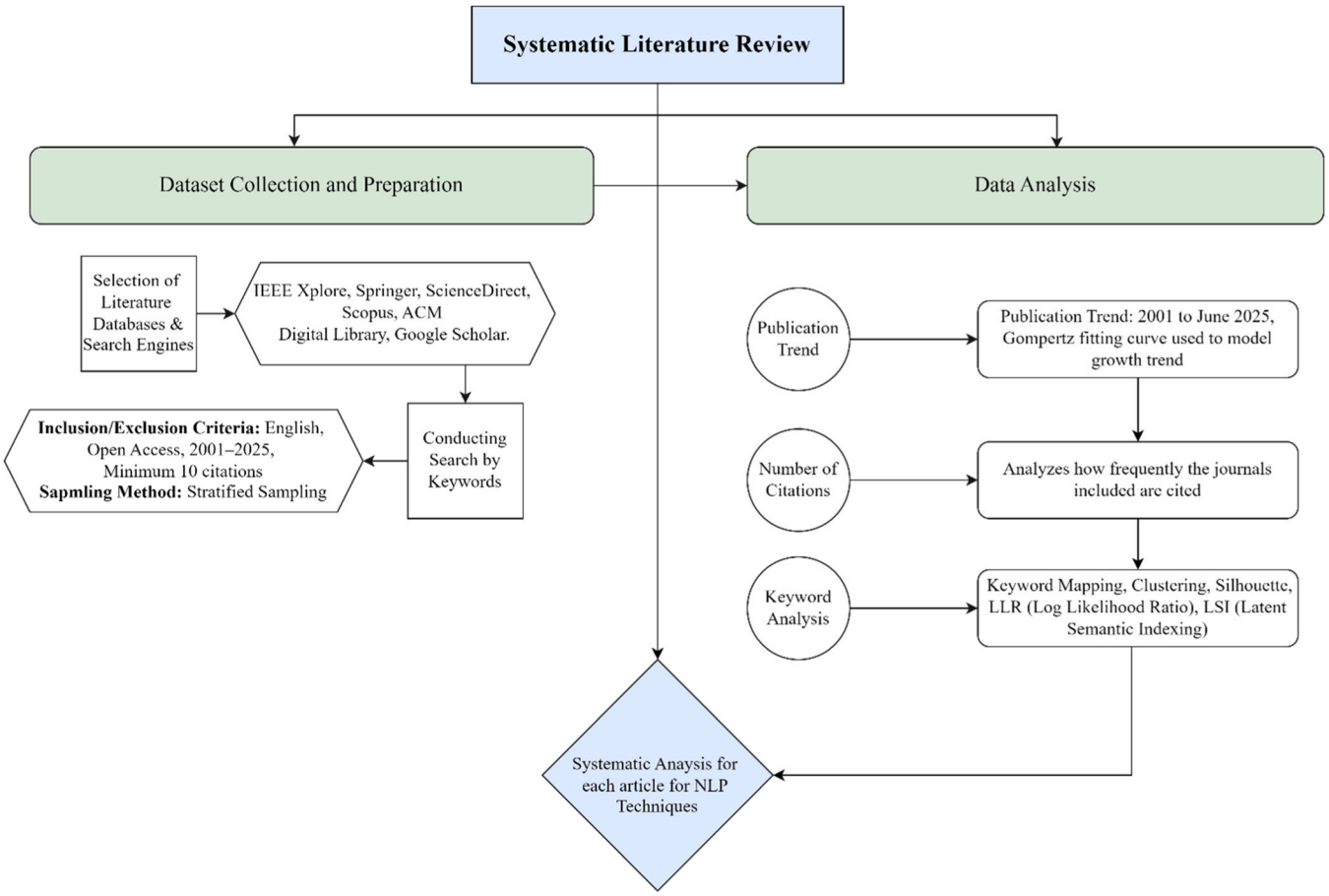


Fig. 1. Methodology adopted for literature analysis.

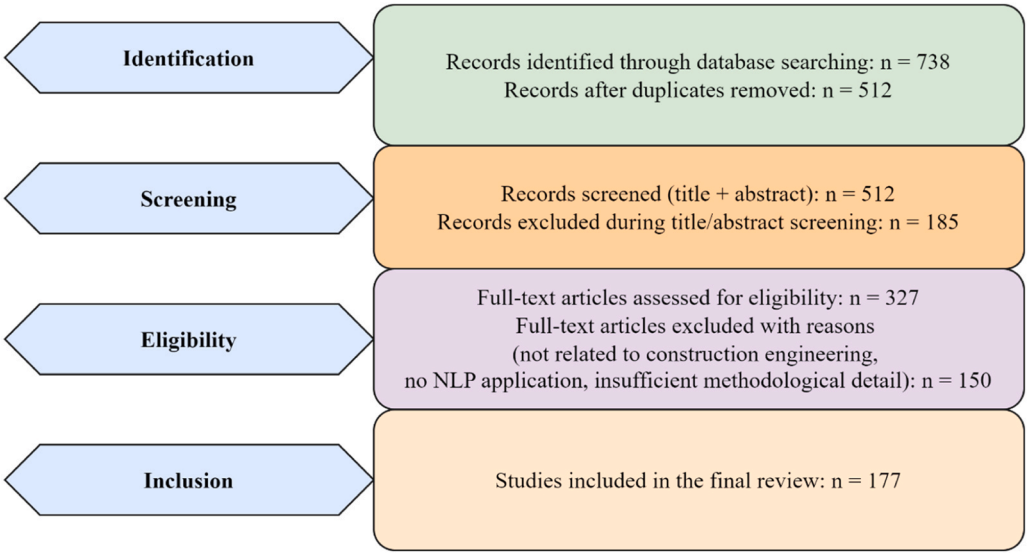


Fig. 2. Prisma Flow Process.

domain or does not have a sufficiently technical approach) were removed at this stage. Therefore, 185 records were eliminated.

At this point, 327 records continued to the Eligibility stage where they underwent detailed examinations of their full texts to verify both their methodological soundness and relevance to NLP applications in construction engineering. Records were removed if they did not apply NLP techniques; do not pertain to construction engineering; or are otherwise technically underdeveloped. At this stage, 150 full-text records were removed from consideration, with documented rationale provided for such exclusion.

Finally, at the end of this multi-stage review process, 177 studies were identified that meet all eligibility criteria and thus were incorporated into the review. Collectively, the 177 studies included in the review represent the current state-of-the-art of NLP-based research related to construction engineering.

2.2. Data analysis

This analysis involved a complete and comprehensive evaluation and synthesis of all of the data and information contained within the selected literature sources. This analysis utilized very basic and easily understood methodologies to obtain accurate and reliable results. A great deal of NLP information relative to Construction Engineering applications was generated by this analysis. This analysis provided both an understanding of the current state of affairs as well as predictive capabilities of NLP in the construction industry. The systematic approach used in this study was designed to provide additional scholarly value beyond what had been provided by previous research efforts and to identify areas of needed additional research relative to the developmental impact of NLP on Construction Engineering.

Construction Engineering research papers related to NLP have grown significantly over the past twenty-four years (from 2001 through June 2025). The growth is an indication that NLP has become increasingly recognized for its ability to transform how construction engineers conduct various tasks, including project automation, risk communications and stakeholder analyses. Early in the 2000s, very little was written about NLP applications in Construction, thus, this area of study was still in development. Since 2015 and particularly since 2020, the number of research studies has grown dramatically; as with other fields, it appears that researchers are now recognizing NLP as a means of facilitating management of unstructured data and improving workflow processes in Construction.

In order to conduct a more thorough analysis of the trend, the data was subjected to a Gompertz fitting curve, as indicated in Fig. 3. The R-square value obtained was 0.934 with confidence level is at 95 % and it signifies that the time duration correlates robustly with number of research papers. An observation of this trend implies a definite pattern in the trend of expansion that has certainly shown correlation. Scholarly articles are steadily increasing, which indicates that it is becoming more accepted that the use of NLP could be employed to address issues in construction engineering. The cases presented in this paper are indicative that the same industry is increasingly approaching the use of NLP methods for handling such tasks as document management, stakeholder communication, information extraction, risk management, and text summarization. Moreover, high value of R square of the observed trend further supports the robustness of the construction engineering domain in terms of research activity and the upward trend of the research activity in NLP.

From 2001–2025, the application of NLP in construction engineering has undergone significant transformation, as illustrated in Fig. 3. In the early years (2001–2014), publication activity remained minimal, reflecting the infancy of NLP applications within the architecture, engineering, and construction (AEC) sectors. Although interest in NLP-based construction research has been building since 2015; this has occurred on a steady basis with the advent of the digital construction era and the increased availability of unprocessed textual data. The dramatic increase in publication rate seen between 2021 and 2025 can be interpreted to indicate an explosive phase of growth as AI, BIM, and demand for analytical intelligence are all contributing factors. Of particular note is the fact that more than twenty publications have appeared in just six months (January 2024 through June 2025) demonstrating an exponential increase in the application of NLP-based research in the construction field. The trendline developed via a Gompertz model indicates a robust predictive path, indicating that while the current pace of NLP-based construction research is accelerating; it will likely eventually plateau due to the maturation of the technology and its eventual full incorporation into construction workflows.

Journal articles and research output are being distributed and cited at different levels and by a number of factors as part of an ever-evolving publication landscape in the NLP application of construction engineering up to June 2025. Engineering has produced the most research papers, totaling 93; followed by Automation in Construction and Journal of Construction Engineering which each had 91 papers and 83 papers, respectively. As the main scholarly publishing outlet for contributions

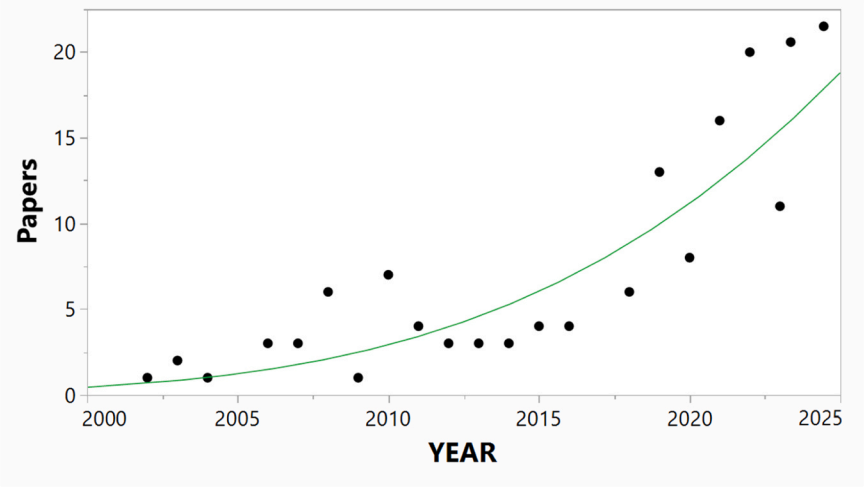


Fig. 3. Number of publications from 2001 to (June) 2025 with Gompertz fitting curve.

that pertain to automation, digitalization, and data-based practices in the built environment, these three journals will likely continue to be key contributors to the body of literature in this area. The ECAM Journal (Engineering, Construction and Architectural Management), has the largest number of citations from the group of surveyed journals and has exceeded 1000 citations, reflecting its increasing academic significance and the relevance of studies it publishes regarding the advancement of the body of knowledge related to intelligent construction management. In addition to Engineering and the Journal of Construction Engineering, Advanced Engineering Informatics and the Journal of Building Engineering have demonstrated high levels of research productivity as well as scholarly influence through their accumulated citations. This indicates that these journals are contributing to the NLP research community in terms of both the quantity of research conducted as well as its scholarly influence. Journals, including Expert Systems with Applications and Construction Innovation, represent new areas within the NLP research community as they draw upon both construction specific communities of AI researchers and broader AI communities. Notably,

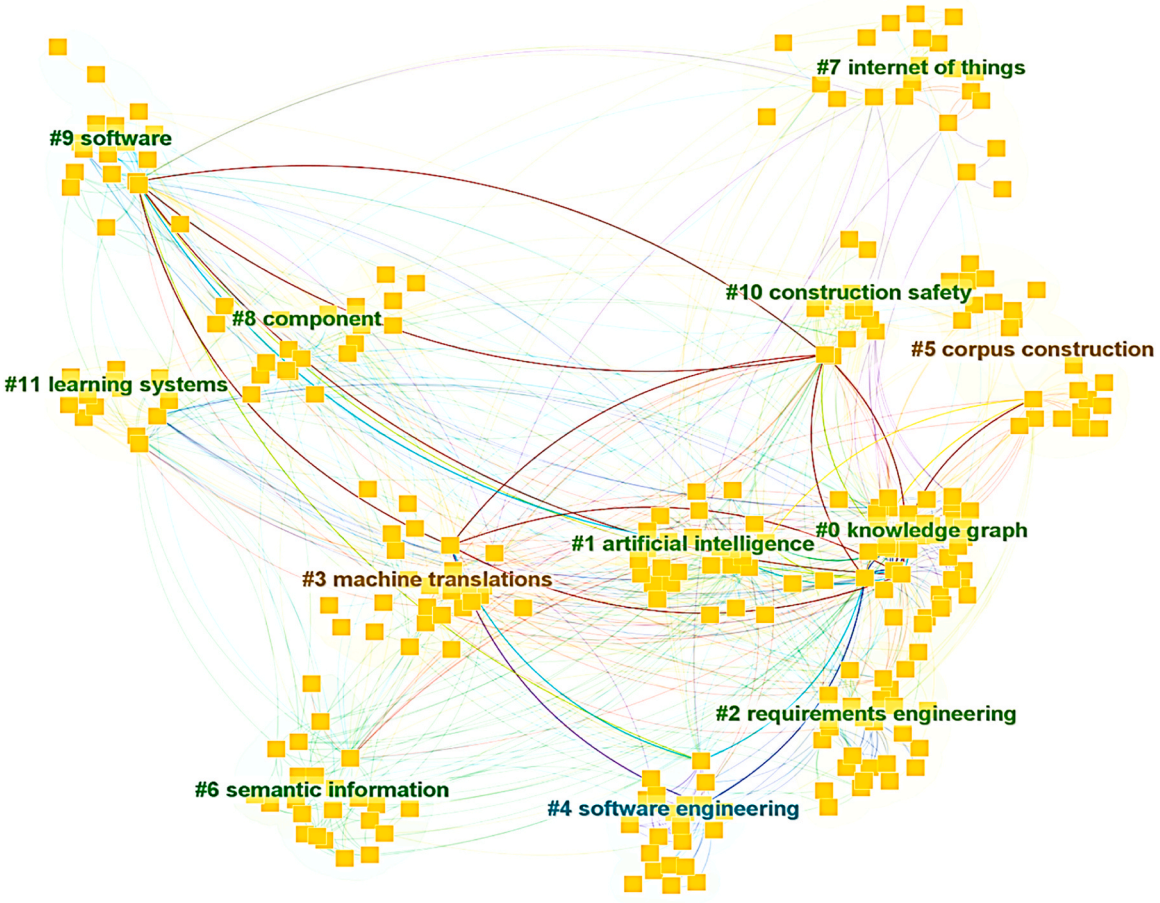


Fig. 4. Network visualization of the keywords.

there are journals that produce large quantities of research but have relatively modest citation counts and vice versa. This highlights that in addition to producing a body of work, the value of the research is also dependent upon how relevant and applicable the work is to the context in which it is being used and therefore, that the scholarly community recognizes and evaluates the research based upon these two criteria. The overall findings of the analysis support the idea that not only is the research field expanding in terms of quantity, but it is also becoming increasingly diverse and deep in terms of the types of research being conducted.

The literature review has identified a collection of keywords that has substantial significance within the domain of NLP as it pertains to the subject of construction engineering, as indicated in Fig. 4. In order to accurately depict the interrelation between these essential terms, the use of Citespace software was utilised to build network visualisations. The use of visualisations in the field of construction engineering may provide valuable insights into the patterns, trends, and linkages present within the research landscape. The found keywords cover a broad spectrum of subjects, providing significant perspectives on the many aspects of study and development within this field.

- **Knowledge Graphs:** Knowledge graphs have usually been viewed as a means to maintain, structure and organise information better, in order to improve retrieval and comprehension efficiency. In the context of construction engineering, where information is abound and intricate, this becomes relevant.
- **Artificial Intelligence:** It represents a strong relationship with NLP and AI based methods in construction field. Further highlights of it are the use of automated document categorization, sentiment analysis and decision support offices.
- **Requirements Engineering:** This implies it contains a theme of understanding and overseeing the various needs and limits within the scope of building, conceivably to construct NLP applications that are more adjusted to the encompassing condition.
- **Machine Learning:** The fact that machine translations exist implies a presence of study domains of the multilingual assistance importance, especially in the global construction industry.
- **Software Engineering:** This means there is some correlation with NLP and different segments of software (which may be related with creation of special software tools and applications for engineering of constructions).
- **Corpus Engineering:** In the realm of construction related activities, it is the process of creating and keeping up specialized linguistic datasets that are directly related to the necessities of training and evaluation of NLP models.
- **Semantic Information:** In the NLP area, specifically for interpreting the importance and context of textual data, it is important. To make precise extraction and analysis of information out of construction documents, it is necessary to comprehend this.
- **Internet of Things:** Its integration suggests a correlation between NLP and IoT applications within the construction industry, potentially including the examination of textual data produced by IoT sensors deployed at building sites.
- **Component & Software:** These are likely associated with the creation and integration of NLP components and software tools specifically designed for applications in the field of construction engineering.
- **Construction Safety:** It is of utmost importance in the construction industry, and NLP the potential to contribute significantly to the analysis of safety-related documents and data, hence improving safety protocols and practises.
- **Learning Systems:** It implies the use of machine learning and artificial intelligence techniques to develop intelligent systems capable of acquiring knowledge from textual material within the domain of building engineering.

This aids researchers and practitioners in developing a more comprehensive grasp of the growing dynamics and areas of attention within the realm of NLP in construction engineering. Fig. 4 presents the network of keywords in accordance with the LLR and the significant links are indicated in red. Through the process of visualising the interconnections among these terms, the study has successfully uncovered new trends, interdisciplinary intersections, and possible research topics that possess the capacity to further propel advancements within the field.

The cluster map in Fig. 5 and Table 1 present overview of the outcomes derived from a clustering analysis performed on a corpus of articles or research papers pertaining to the domain of NLP within the context of construction engineering. In Fig. 5a total of 12 clusters, numbered from 0 to 11, are presented. The clusters and links between clusters is mainly based on five colors (Cyan, Blue, Green, Yellow, and Red), presenting the overall relationship significance of the clusters with respect to similar content based on keywords. The table has separate clusters of documents, with each row representing one cluster. From this summary, numerous significant observations may be derived. First and foremost, it is apparent that the clusters exhibit variations in size, with some clusters including a greater quantity of documents compared to others. This observation indicates that certain domains, specifically those related to NLP and Building Engineering, have been subjected to greater scholarly study and/or have generated a larger volume of published research. The silhouette scores, which can be considered a measure of the quality of the clusters produced by the clustering technique, indicate that a large number of clusters are characterized by very high silhouette scores. As such, it appears that a large proportion of documents within each cluster appear to contain similar content and themes. The fact that a large number of clusters exhibit a high level of similarity also represents one possible way to assess the success of the clustering methodology in effectively grouping like documents. The mean year value in the column labeled "Mean Year" presents the average year at which documents were published for each cluster. Due to the wide time frame over which studies on the application of NLP in the field of Construction Engineering have been conducted, i.e., from 2005 to the present day, with regard to the most recent publications on the subject, it may be inferred that there is a strong indication of ongoing innovation and development in this area of study. Additionally, the data presented in the columns labeled "Top Terms (LSI)" and "Log Likelihood Ratio (LLR)" provide information regarding the terms and concepts associated with each cluster. These terms represent indicative elements of the theme(s) contained within each cluster of documents and therefore provide insight into the specific topics and areas of focus represented by the studies encompassed by this study.

In response to recent developments this review methodology is extended to incorporate literature related to Reinforcement Learning with Human Feedback (RLHF) and evaluation frameworks for large language models (LLMs). These emerging themes were integrated into the review protocol to better capture evolving trends in human-AI collaboration, interpretability, and benchmarking of NLP systems in construction. Studies meeting the inclusion criteria were re-evaluated to identify contributions addressing RLHF applications, LLM alignment techniques, and benchmark-based model performance assessment, ensuring a more comprehensive and forward-looking analysis.

3. Understanding of NLP

3.1. AI-driven text analysis techniques

NLP forms the basis for text analysis techniques driven by Artificial Intelligence (AI). NLP encompasses a diverse array of methodologies and algorithms that have been specifically developed to facilitate the ability of computers to comprehend, interpret, and generate human language [24,25]. In general, the task of analyzing text is known as a natural language understanding (NLU) task in the world of AI. The field of study

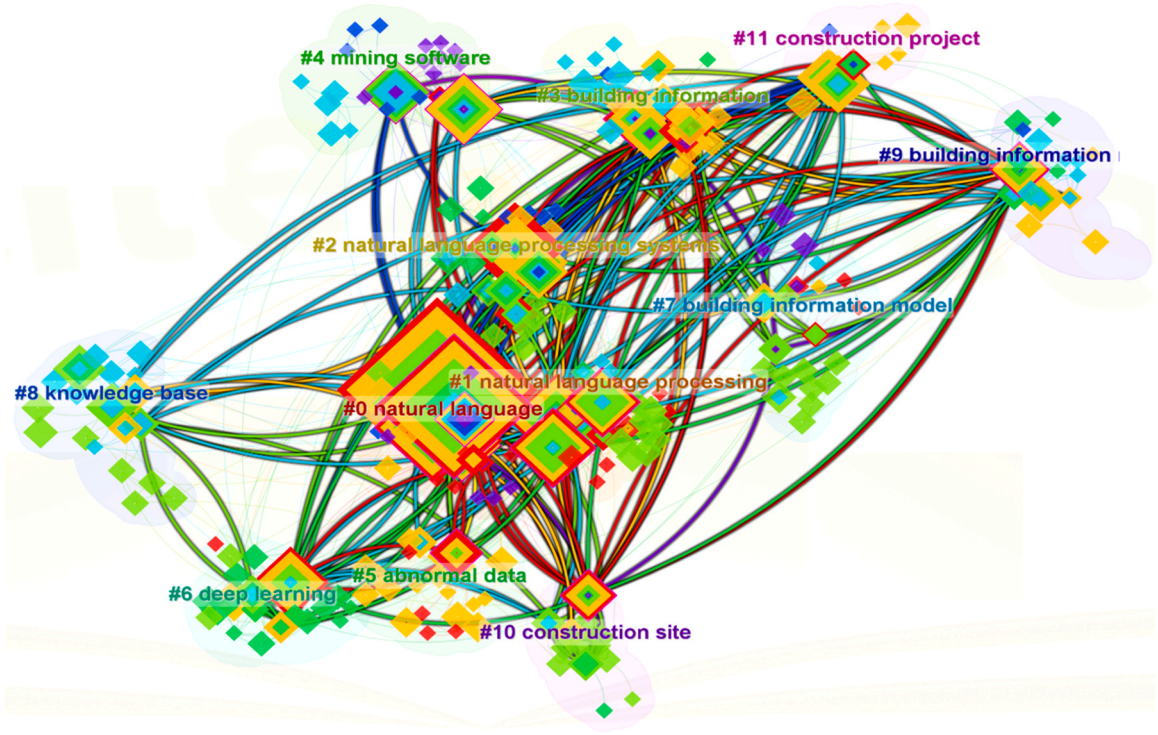


Fig. 5. Cluster map of keywords.

called NLU aims at the development of techniques and algorithms that are able to make machines understand and interpret human language [26,27]. Therefore, this can help to improve conditions of the decision making procedures and increase the level of the project's efficiency [28,29]. The technique of topic model is widely recognized in the area of AI, and has been received considerable attention in scholarly literature [30]. Since topics indicate often recurring themes and thus patterns that appear in different documents, topic modeling techniques are able to find topics that summarize issues or challenges related to a certain project [31,32]. Named Entity Recognition (NER) is the technique used in text analysis to locate and classify such distinct entities as a project stakeholder, location, a material or an equipment, in a given body of text [33,34]. NER in construction organisations has the potential to save a great deal of time and resources by automating the process of project data manual extraction and associated organisation [35]. By looking at the communication channels of a project, for example, emails, social media posts or customer feedback, sentiment analysis can derive an impressive insight into the attitudes and the perceptions of stakeholders [36,37]. Reduction in length of a text by generating shorter yet comprehensive summaries of the input text that preserves the important information and key ideas of the input is known as text summarization [38]. These aforementioned studies and other relevant research in the field also indicate the promising prospects of utilization of artificial intelligence based text analysis methodologies in this specific area of application: construction engineering [39,40].

More recently, reinforcement learning approaches have been adopted to optimize query formulation and retrieval in construction information systems. These models learn from user interactions and document relevance scores to iteratively refine search queries, improving semantic retrieval performance. This is particularly useful for engineers seeking specific clauses, design changes, or RFI threads in large digital archives. By automating and personalizing the search process, these systems reduce information overload and ensure that critical insights are surfaced more efficiently during design coordination or site reporting [30,41].

One of the most important advancements in using AI for analyzing text is Reinforcement Learning with Human Feedback (RLHF). RLHF works at the intersection of supervised learning and improving interactive models that work with domain experts; so, historically, models have been trained based on the past, while models can now be trained on real-time feedback and expertise from experts working in construction [42,43], which is particularly well-suited for construction because construction has many areas where interpretability is required, there is much legally sensitive information, and safety-critical language is used often. RLHF can improve the reliability of the extraction, classification, and summarizing of technical documents that are created by language models, and will improve the level of both accuracy and trustworthiness of construction NLP systems, and therefore represents a significant change in how AI models evolve in high-risk environments [44,45].

3.2. *Characteristics of NLP*

The core characteristics of NLP from the perspective of its application in the construction engineering are as follows:

- **Ambiguity Handling:** Ambiguity management represents a significant hurdle in the field of NLP, primarily due to the intrinsic ambiguity present in human language. This challenge is being tackled through the use of the algorithms and the models of NLP, which out of the box have the capacity to disambiguate and interpret the language in the content where the focus is on the use of the language inside its context [46,47]. These techniques are employed to effectively address the issue of ambiguity [48,49].
- **Contextual Understanding:** The comprehension of language is heavily influenced by its context, and NLP systems strive to capture and exploit contextual information in order to improve their understanding [50]. One major area for NLP has been in devising various techniques for capture of contextual information to boost the accuracy of the models [51]. Examples of such techniques include

Table 1
Summary of the identified clusters.

Cluster	Size	Silhouette	Mean Year	Top Terms (LSI)	Log Likelihood Ratio (LLR)
0	42	0.748	2021	natural language; empirical engineering knowledge; long term evolution; knowledge base; cnc equipment | NLP; knowledge base; cnc equipment; text analysis; term co	knowledge graph (34.5, 1.0E-4); text mining (27.55, 1.0E-4); knowledge graphs (20.6, 1.0E-4); knowledge engineering (18.35, 1.0E-4); artificial intelligence (14.96, 0.001)
1	27	0.879	2010	natural language; NLP; artificial intelligence; machine learning; building information modeling | engineering procurement; risk assessment; javascript object notations; risk analysis; con struction	artificial intelligence (35.51, 1.0E-4); machine learning (32.21, 1.0E-4); phrase matcher (19.95, 1.0E-4); risk phrase extraction (15.95, 1.0E-4); risk assessment (11.96, 0.001)
2	26	0.798	2009	natural language; NLP systems; physical system; abstracting; graph theory | semantics; pipelines; grammatical frameworks; data driven approach; grammar engineering	requirements engineering (24.45, 1.0E-4); software requirements (13.21, 0.001); software design (12.17, 0.001); pipelines (8.95, 0.005); information use (8.95, 0.005)
3	26	0.655	2005	natural language; building information; NLP; building information modeling; artificial intelligence | semantic web; project information; construction industry; bidirectional encoder representation; systematic mapping	machine translations (14.93, 0.001); building information modeling (10.87, 0.001); building information modelling (10.79, 0.005); machine translation (10.6, 0.005); rule based (9.94, 0.005)
4	23	0.88	2014	natural language; mining software; bug report; nsga ii; human computer | topological functioning; NLP systems; nsga ii; human computer; graphical user interfaces	software engineering (25.7, 1.0E-4); topological functioning model (12.04, 0.001); topology (12.04, 0.001); nsga ii algorithm (10.93, 0.001); controller synthesis (10.93, 0.001)
5	23	0.865	2021	natural language; abnormal data; electric power transmission; smart power;	corpus construction (14.73, 0.001); code language model (12.75,

Table 1 (*continued*)

Cluster	Size	Silhouette	Mean Year	Top Terms (LSI)	Log Likelihood Ratio (LLR)
				software development | source code; source code pre; zero shot; open source; software development	0.001); abnormal data detection (12.75, 0.001); fake news (12.75, 0.001); anomaly detection (12.75, 0.001)
6	23	0.808	2020	natural language; NLP; deep learning; construction site; structural health | long distance; long short term; regulation information; construction procedure relation; graph neural	semantic information (16.29, 1.0E-4); random processes (11.93, 0.001); attention mechanisms (10.85, 0.001); long distance dependency (10.85, 0.001); hydraulic engineering (10.85, 0.001)
7	22	0.824	2006	natural language; NLP; building information model; digital twin; construction engineering | speech recognition; automatic speech; industry foundation classes; human machine; economic education	internet of things (21.24, 1.0E-4); augmented reality (19.6, 1.0E-4); building life cycle (13.05, 0.001); speech recognition (13.02, 0.001); building information model bim (13.02, 0.001)
8	20	0.835	2021	natural language; knowledge base; word sense; neural networks; chinese natural language | NLP; automatic question; bidirectional long short term; long short term memory; NLP systems	component (18.38, 1.0E-4); knowledge base construction (17.2, 1.0E-4); knowledge based systems (15.78, 1.0E-4); contacted features (11.61, 0.001); trigger words (11.61, 0.001)
9	18	0.765	2020	natural language; NLP; building information modeling; bidirectional encoder representation; machine learning | web services; industry foundation classes; active server; information management; bidirectional encoder representation	software (18.68, 1.0E-4); architectural design (16.39, 1.0E-4); aec industry (12.44, 0.001); digital architecture (12.44, 0.001); computational architecture (12.44, 0.001)
10	13	0.859	2021	natural language; deep learning; NLP; construction site; structural health | automatic information; cross combination; construction accident news; small sample; large scale	construction safety (16.69, 1.0E-4); incident (11.11, 0.001); automatic information extraction (11.11, 0.001); intelligent manufacturing (11.11, 0.001); association rule

(*continued on next page*)

Table 1 (*continued*)

Cluster	Size	Silhouette	Mean Year	Top Terms (LSI)	Log Likelihood Ratio (LLR)
11	13	0.876	2010	natural language; NLP; construction project; construction site; symbiotic organisms | source code; source code pre; zero shot; open source; text analysis	mining (11.11, 0.001) learning systems (14.2, 0.001); accurate (11.91, 0.001); engineering education (11.13, 0.001); tourism management (9.88, 0.005); neural network (8.2, 0.005)

word embedding, contextual word representations (BERT, GPT, etc.) [52,53].

- **Language Generation:** NLP encompasses not only the comprehension and processing of human language but also encompasses the generation of responses that are coherent and contextually suitable. Therefore, with the aim of producing outputs that resemble other human language as much as possible, NLP models like language models, and neural machine translation models have been devised. Language generation capabilities make possible the implementation of several different applications such as chatbots, virtual assistants, as well as the ability to generate content automatically [54,55].
- **Multilingualism for International Projects:** The phenomenon of multilingualism is characterized by a significant level of linguistic diversity, as evidenced by the existence of numerous languages spoken across various regions of the globe. They include a number of linguistic communities, such as tokenization, segmentation, language identification, and machine translation, etc [56]. They try to make it easier to transfer a model or a piece of knowledge from one language to another to avoid needing language specific training data and resources [57,58]. More recently, NLP researchers have developed multilingual transformer models tailored to construction project workflows involving cross-border collaboration.
- **Adaptability and Learning:** Adaptability and learning are two characteristics that are at the base of the NLP (Natural Language Processing) systems. NLP systems have been able to use machine learning algorithms and deep neural networks to find patterns and relationships in large training datasets through a process of learning from those large datasets. Through using supervised learning, unsupervised learning and reinforcement learning methods, NLP systems can learn to accomplish specific tasks, such as sentiment analysis, named entity recognition, machine translation etc. [59,60].

Alongside the developing architecture of NLP techniques, we are observing increasing interest in mechanisms that support model alignment with expert judgments in high-risk domains including construction. One potential avenue for this development includes Reinforcement Learning with Human Feedback (RLHF), which enables large language models (LLMs) to modify their output responses based upon an individual's preference for their response. RLHF can enhance transformer-based models by incorporating a feedback loop that iteratively assesses and fine-tunes performance [61,62]. As general-purpose LLMs may interpret or represent domain-specific construction information inaccurately or ambiguously, RLHF will enable NLP to not only be more accurate, but also safer and more transparent when applied to real-world applications including hazardous condition identification, technical compliance assessments and specification interpretation.

3.3. Factors affecting NLP application in construction engineering

The Fishbone Diagram, also referred to as the Ishikawa or Cause-and-Effect diagram indicated in Fig. 6, serves as a visual representation of the many interconnected aspects that exert effect on NLP within the domain of construction engineering. This study will focus on identifying the main subject which is found to be "Factors Influencing NLP in the Field of Construction Engineering." There are four primary branches that come out from this centre point which bounds a unique classification of impacts [63–65]. The first branch of study is 'Technology and Data Integration', which first dilves into NLP technology and then the problems and advancement of integrating different data sources to build a project. Finally, the fourth branch “Ethical Considerations and Stakeholder Engagement” deals with the ethical aspects of NLP in the construction industry and the importance of NLP in successful stakeholder engagement among others [48,49]. Collectively, these branches collectively provide an all-encompassing visual depiction of the many elements influencing the domain of NLP in the field of construction engineering. This framework facilitates thorough examination and offers guidance for future research endeavours.

3.4. Benefits of NLP in construction engineering

NLP has the capability to automate labor-intensive manual tasks, such as the categorization of documents and the extraction of information. It additionally permits the professionals to believe the ruminations of the stakeholders. And in the area of regulatory compliance, NLP is used to automate spotting of clauses that are non-compliant, deviations or omissions of clauses in information written down in project documentation [66,67].

NLP has advantages for construction that include better document management, better risk management, better informed decision making and knowledge management, better communication and stakeholder engagement, and better-quality control and compliance monitoring. By utilizing NLP techniques, construction firms have the potential to enhance their operational procedures, enhance project results, and attain a competitive advantage within the industry [68,69]. Following are the most critical benefits found from literature:

- **Improved Document Management and Information Retrieval:** Comprehensive documentation for all construction projects is a necessity (project plans, specifications, contracts, etc.) for improved documentation management & information searchability; additionally, NLP may enable construction professionals to utilize semantic search functionality to find relevant information based on semantic meaning rather than using only an exact match of keywords [70,71].
- **Enhanced Risk Management:** Risk identification and risk management are major contributors for better Risk Management Practices in Construction Projects. NLP could also contribute to improved risk management practices via the review of project related documents and communication. All Construction Organizations can utilize NLP techniques (topic modeling, sentiment analysis) on project related documents and records (contracts, project reports, communications etc.) to identify potential future risks, assess stakeholders' views and undertake corrective action as needed [72,73].
- **Improved Decision-Making and Knowledge Management:** Algorithms for NLP can automatically find patterns, trends and anomalies in project documentation and communication as they are analyzed by an NLP algorithm. The use of NLP techniques provides the opportunity to analyze information concerning project-related data and thus facilitates the dissemination of the information and the knowledge exchange among team members. [74,75].
- **Enhanced Communication and Stakeholder Engagement:** Effective Communication and Stakeholder Engagement is important for the success of Construction Projects. Sentiment Analysis and Named Entity Recognition (NLP Techniques) can be used to provide valuable insights to Construction Organizations regarding the opinions, sentiments, and perceptions of stakeholders. The information provided will allow the organizations to adjust their communication strategy,

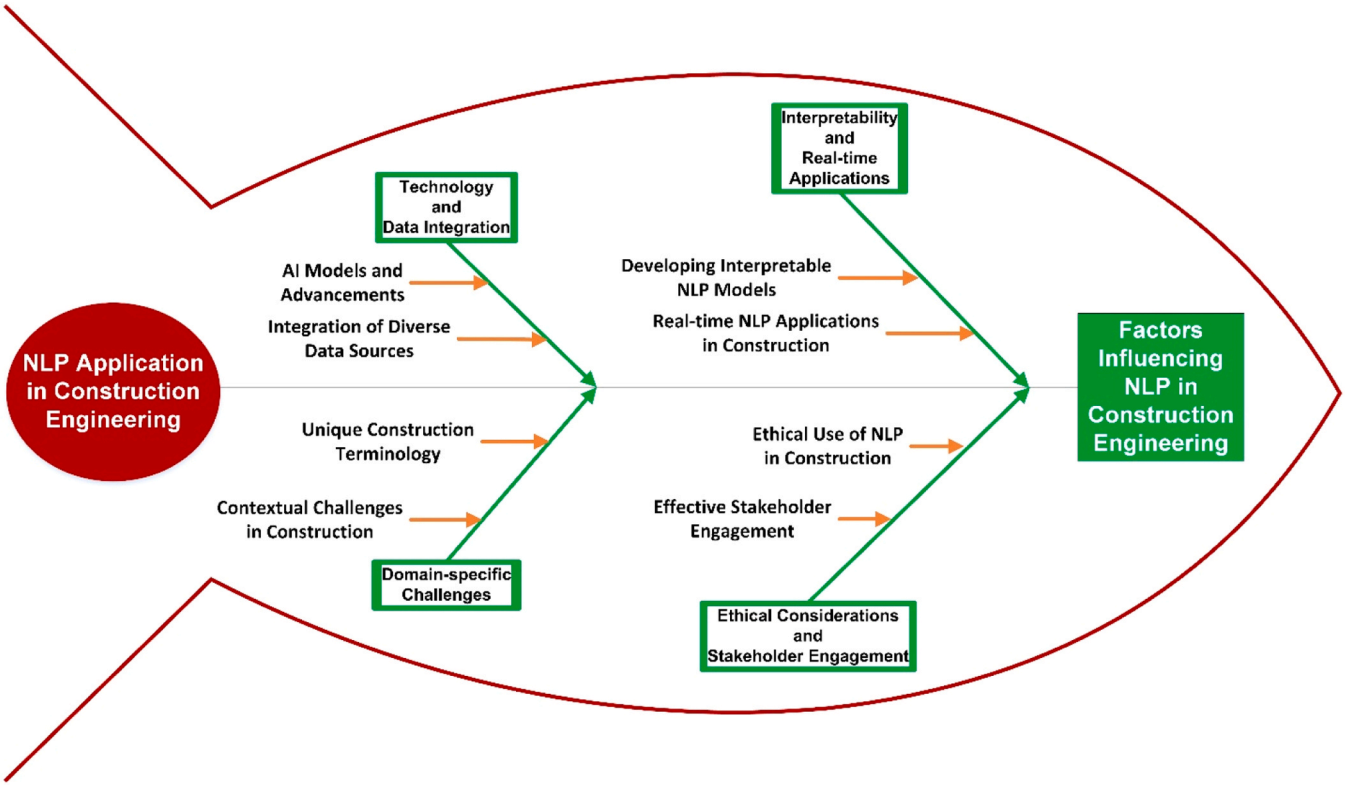


Fig. 6. Fishbone Diagram of Factors Influencing NLP in Construction Engineeri.

address issues that are of concern to stakeholders, and create more effective engagement; thus allowing for stronger relationship building between stakeholders and the organization, mitigating conflict, and increasing overall satisfaction of all parties involved in the project [76,77].

- **Automated Analysis of Contractual Claims:** Research conducted recently on the application of NLP to automate the analysis of construction contracts, claims and dispute documentation, have identified how Named Entity Recognition (NER) and Dependency Parsing could be used to identify common claim patterns, reasons for delays and ambiguous contractual terms across multiple documents of a large scale construction project [78]. Through streamlined claims review, improved transparency, NLP also aids legal teams and contract managers mitigate disputes and speed up timeframes for resolving disputes. In particular, this will be beneficial in the case of major infrastructure projects where the volume of documentation is extremely large, and there is a high potential for litigation [79].
- **Efficient Quality Control and Compliance Monitoring:** The construction sector is heavily dependent on the implementation of quality control and compliance with regulatory requirements. Quality issues can be identified using NLP through analysis of textual data within, inspection reports, defect logs, and customer feedback. The construction sector is heavily dependent on the implementation of quality control and compliance with regulatory requirements. Quality issues can be identified using NLP through analysis of textual data within, inspection reports, defect logs, and customer feedback [80,81].
- **RLHF:** Because NLP output is so much dependent on both the quality of legal terms and safety-related information in construction documents, the dependability of NLP output is an essential element. The Reliability of LLM Outputs has been a focus of research and development using RLHF to provide domain experts' input in either the training phase of the LLM or fine-tune phase of the LLM. RLHF can help ensure the LLM's interpretation of data meets legal requirements and industry standards when used in applications such as identifying clauses in contracts or assessing safety risks. If not done, LLMs could produce content that either ignores critical elements or overstates potential risks. Therefore, RLHF can be incorporated in construction NLP applications to increase the reliability and accountability of the applications [43,44,82,83].

4. Research topics of NLP in construction engineering

The subsequent five research topics, indicated in Fig. 7, in NLP hold substantial importance within the domain of Construction Engineering. Sentiment Analysis in Stakeholder Communication within Construction Projects investigates various methodologies for sentiment analysis in order to assess stakeholder communication. The process of extracting essential information from construction project documents entails the utilization of NLP techniques. These techniques are employed to extract crucial information from unstructured textual data, which includes project requirements and specifications. The application of NLU in the field of Construction Risk Management focus to improve the management of risks in construction projects. This is achieved through the utilization of NLP techniques, which involve the analysis of various project-related documents and communication.

4.1. Automated classification and categorization of construction project documents

Numerous studies have been conducted to examine the utilization of NLP techniques in the context of document classification and categorization within the construction industry. Shuai [14] developed a hybrid model which combined machine learning approach, and NLP techniques in order to automate the classification of construction project documents [84,85]. In analogy, the researchers Demertzis et al. [1] developed a document categorization system of documents related to construction safety management. In their study, the researchers used part of the

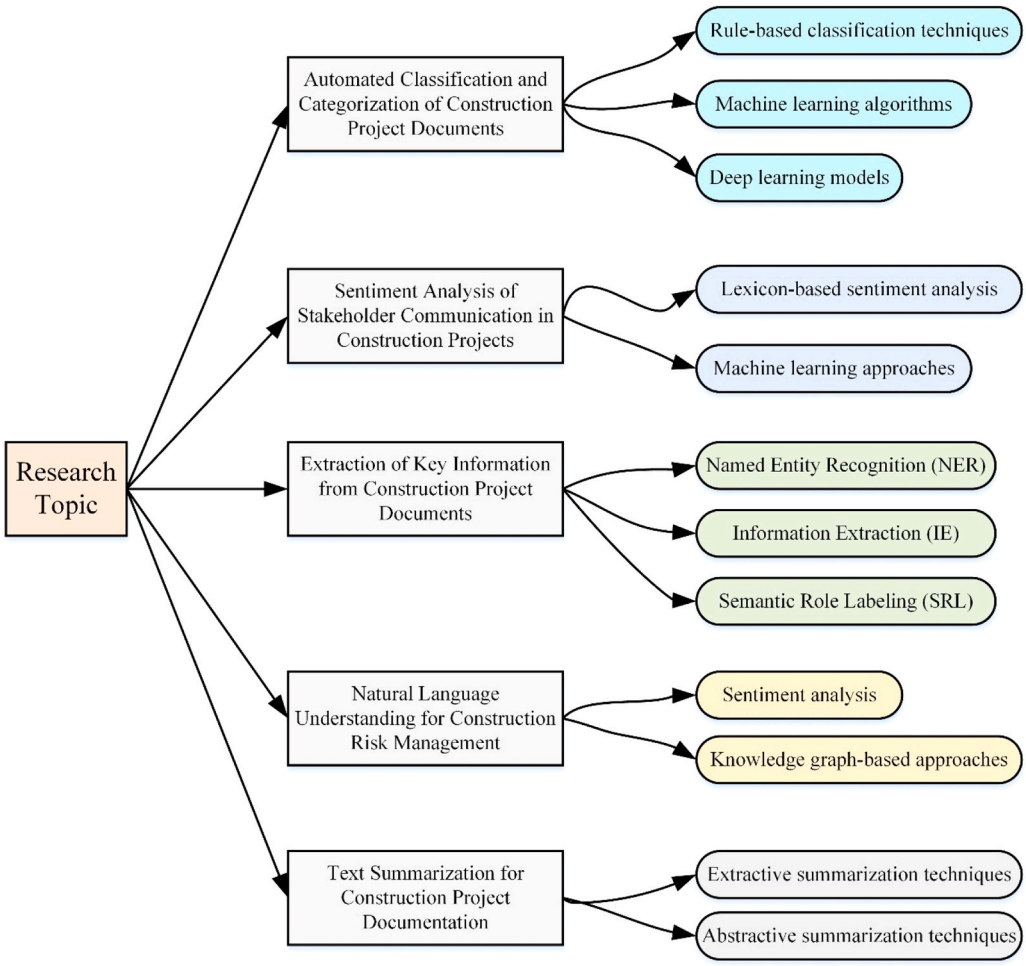


Fig. 7. Summary of main research topics and their relevant AI approaches.

speech tagging and named entity recognition to preprocess textual data [86].

One of the problems to the automation of document classification and categorization is the lack of standardised schemes for building project documents. The process of building a universal classification system is very difficult, because of the nature of the building projects and large volume of document types and formats [87]. An example of this may be seen in the work of Yin et al. [19], where they introduced a categorization framework for construction documents that included domain-specific information and criteria.

Deep learning architectures have also demonstrated strong performance in text-classification tasks beyond the construction domain, such as the detection of depression and suicidal tendencies in social-media data [88], highlighting the generalisable potential of feature-optimized NLP pipelines.

Table 2 is showing NLP has advantages for construction that include better document management, better risk management, better informed decision making and knowledge management, better communication and stakeholder engagement, and better quality control and compliance monitoring [104]. An example of deep learning automation for building project document categorization has been published by Chowdhury and Alzarrad [4]. They used methodology consisted of integrating convolutional neural networks (CNNs) to extract features from document pictures along with recurrent neural networks (RNNs) to capture effective temporal relationships in textual data [105].

4.2. Sentiment analysis of stakeholder communication in construction projects

Many studies have investigated the application of Sentiment Analysis methodologies for understanding Stakeholder sentiment (opinions) in Construction project contexts. Many studies have investigated the application of Sentiment Analysis methodologies for understanding Stakeholder sentiment (opinions) in Construction project contexts. Then using the Tovar Onofre & Camargo [15] sentiment analysis was applied on the social media data to determine the general common sentiments of people about the construction projects. This allowed project managers to watch public views rapidly, and thus rapidly answer any concerns that

Table 2
Relevant studies for Automated Classification and Categorization of Construction Project Documents.

Method	Purpose	Reference
Rule-based classification	Automate document classification and categorization process	[89–91]
Machine learning (ML) algorithms	Improve accuracy and efficiency of document classification	[92–94]
Deep learning (DL) models	Handle complex construction project documents	[95–97]
NLP techniques	Extract key information from unstructured textual data	[98–100]
Hybrid approach (ML and NLP)	Combine ML and NLP techniques for better classification	[101–103]

arose [106].

In a similar vein, Saka et al. [107] employed sentiment analysis methodologies to examine customer feedback data within the construction sector. An essential obstacle encountered in the analysis of sentiment in stakeholder communication within construction projects pertains to the discernment of words and phrases that convey sentiment within the context of the construction domain. In order to tackle this particular challenge, scholars have undertaken investigations into the creation of sentiment lexicons that are specifically designed for construction-related language [108,109].

An example of such kind is the work of Saka et al. [8], they created the sentiment lexicon for the domain construction project management. The use of a domain specific sentiment lexicon improved the precision of sentiment analysis, since it covered the fine grained and context dependent interpretations of the terms relating to construction. A wide variety of data communication channels are present in construction projects, including textual data, visual representation, video and audio recording. In order to obtain sentiment information from various communication channels, scholars have conducted investigations into the integration of text analysis with image analysis and video analysis and with audio analysis [110,111].

In the education of sentiment in construction project feedback, Shepitko et al. [9] proposed a framework for multimodal sentiment analysis. This advancement has provided the construction organization with more profound understanding of the stakeholder's sentiments [112, 113]. Building on this, recent studies have applied sentiment analysis techniques to post-occupancy evaluations and facility management feedback. By mining client reviews, maintenance reports, and tenant comments, NLP models can identify common pain points such as thermal discomfort, accessibility issues, or service delays. Moreover, combining aspect-based sentiment analysis with facility usage logs helps organizations align future design choices with user experience metrics [114,115]. Most of the sentiment analysis can be enhanced using sophisticated construction language machine learning and deep learning models. By incorporating domain expertise, it helps identifying project specific sentiment patterns as well as interpreting sentiment score from the lens of industry specific factors (refer Table 3).

4.3. Extraction of key information from construction project documents

A multitude of scholarly investigations have been conducted to examine the utilization of NLP techniques for the purpose of extracting information within the construction industry. According to Campos-Medina et al. [5], a framework for information extraction on the construction project specification was developed. The framework had higher levels of accuracy and efficiency than manual extraction method that used in the conventional way. In a similar vein, Jacques de Sousa, Poças Martins, & Sanhudo [2] put forth a NLP oriented methodology to extract requirements from documents pertaining to construction projects. The approach demonstrated favorable outcomes in effectively discerning and extracting project requirements, resulting in a reduction of the time and effort needed for manual extraction.

Hybrid approaches use rule based, pattern based, and machine learning techniques all together so as to augment the precision and resilience of information extraction models. It is possible to develop models of construction documents that can capture the semantics and interdependencies there, and the use of such models can aid more accurate extraction of important information. Analyzing and extracting information from visual data like construction drawings or site images may be one useful method to complement the building information [131,132] (refer Table 4).

4.4. Natural language understanding for construction risk management

The researchers employed machine learning algorithms, specifically Support Vector Machines (SVM), to categorize project-related documents based on risk and extract pertinent risk-related information. The framework exhibited favorable outcomes in enhancing the precision and effectiveness of risk identification procedures [94,143]. In a similar vein,

M Locatelli et al. [34] For the purpose of examining project documents and predicting potential risk, proposed a risk prediction based on NLP techniques to integrate. In order to extract risk related information from textual data, the researchers used topic modeling and sentiment analysis and constructed risk prediction models. Construction projects encompass a diverse array of textual resources, such as project plans, specifications, contracts, and records of communication [145].

Saini et al. [39] introduced a framework for extracting risk knowledge from construction project documents, employing NLP techniques. Named entity recognition and topic modeling methods were used to spot and categorize terms and concepts of risk in the studies [36,95,120]. To address this specific challenge, scholars have been investigating into the ontology and lexicon creation which are specialized to specific domains [146]. An example of this is the work by Shen et al. [50], who had created an ontology for construction risk by crunching together domain-specific expertise with NLP paradigms [147,148]. By using construction risk ontology, comprehensibility of risks was enhanced; taxonomy and control breakdowns of risks can be enhanced to identify the risks; proactive risk management was achievable. [149]. Furthermore, future research in the field of natural language understanding for construction risk management can be further investigated on several veins of research employing sophisticated deep learning architecture like recurrent neural network (RNN), or even transformers in risk related information extraction and analysis domain can further improve the precision and contextual understanding of risk related textual data. Use of NLP techniques and historical project data in designing risk prediction models can allow proactive risk man-agement and aid in decision making process [150,151].

Among recent developments, several studies have introduced construction-specific knowledge graphs that semantically map relationships between safety incidents, root causes, site conditions, and mitigation strategies. These knowledge graphs are generated by mining accident reports and linking entities such as locations, equipment, and worker roles to specific risk categories. The structured representation

Table 3
Sentiment Analysis of Stakeholder Communication in Construction Projects.

Method	Purpose	Reference
Lexicon-based approach	Analyze sentiment in stakeholder communication	[116–118]
Machine learning (ML) algorithms	Classify sentiment in stakeholder communication	[119–121]
Hybrid approach (Lexicon and ML)	Combine lexicon-based and ML techniques for sentiment analysis	[122–124]
Deep learning (DL) models	Capture nuanced sentiment in stakeholder communication	[125–127]
Aspect-based sentiment analysis	Identify sentiment towards specific aspects in stakeholder communication	[128–130]

Table 4
Extraction of Key Information from Construction Project Documents.

Method	Purpose	Reference
Named Entity Recognition (NER)	Extract key entities and information from project documents	[133–135]
Information Extraction (IE)	Automatically extract specific project information from textual data	[136–138]
Semantic Role Labeling (SRL)	Identify the roles and relationships of key information in documents	[139–141]
Topic modeling	Discover latent topics and extract key information from documents	[142,143]
Deep learning (DL) models	Learn representations of key information for extraction tasks	[135,137, 144]

enables enhanced visualization and querying of risk propagation paths, supporting proactive decision-making and training interventions. When integrated with real-time site data, such systems can also serve as dynamic safety monitoring platforms [152,153] (refer Table 5).

4.5. Text summarization for construction project documentation

A multitude of scholarly investigations have been conducted to examine the utilization of NLP techniques for the purpose of text summarization within the realm of construction. A study [25] developed a summarization system specialized for construction domain as well as employs NLP techniques to create summaries from project documents as shown in Table 6. The system worked more efficiently and effectively in the process of summarizing information about construction projects [10]. A study [48] also proposes a method to summarize construction project documents using fusion of NLP techniques and machine learning algorithms. Favorable outcomes of the proposed methodology in producing short and informative summaries of construction project documents.

Scholars have conducted investigations into the advancement of domain-specific lexicons and ontologies in order to more effectively capture the language specific to a particular domain and enhance the precision of text summarization [48,128,160]. Singh et al. [46], presented an example of using ontology in the construction domain by developing a construction ontology with the aim that of helping in constructing a summarization process for construction project documents. The use of domain specific ontologies has the potential of producing summaries are more pertinent and comprehensible.

Construction projects encompass a diverse range of documents, such as project plans, specifications, contracts, and reports, each exhibiting distinct structural and formatting characteristics. To address this issue, Mah et al. [35] proposed a new construction project document hybrid summarization approach that combines extractive and abstractive summarization. Neural network based models for abstractive summarization was also employed where they generated summaries by paraphrasing and rephrasing the extracted sentences [46,122,161]. Using a transformer model (BART/T5), more recent research has built upon the above foundation for use in abstractive summarization of multiple types of documents used in complex construction projects (i.e., change orders, progress reports, technical meetings). The new generation of transformer-based models have been able to provide better results than previous models based on their ability to preserve context when summarizing, and are now being used to summarize complex technical documents into executive-level summaries that capture critical information about each project. As such, these models will be most beneficial to project stakeholders that require periodic summaries of project activities and would like to avoid the time-consuming process of reviewing voluminous amounts of project documentation [162].

Potential future research directions within the field of text summarization for construction project documentation can involve a variety of investigations. In addition, the use of different types of data modalities (i.e., image/diagram representations as well as textual) could also potentially enhance the overall comprehension of the construction project documentation. Developing specialized evaluation frameworks and datasets for assessment of construction project documentation will assist researchers in reliably assessing the effectiveness and performance of text summarization methodologies [8,107,163] (refer Table 6).

Table 5
Natural Language Understanding for Construction Risk Management.

Method	Purpose	Reference
Sentiment analysis	Analyze stakeholder sentiments related to risk in construction projects	[39,119, 130]
Text classification	Classify project documents based on risk-related content	[40,154, 155]
Knowledge graph	Build a semantic representation of risk-related information	[25,131, 141]
Information retrieval	Retrieve relevant risk-related information from textual data	[146,149, 156]
Risk pattern recognition	Identify patterns and trends in risk-related information	[136,157, 158]

Table 6
Text Summarization for Construction Project Documentation.

Method	Purpose	Reference
Extractive Summarization	Extract key sentences to summarize construction project documentation	[57,99, 113]
Abstractive Summarization	Generate concise summaries by understanding the context and generating new sentences	[6,7,10]
Reinforcement Learning	Train models to generate summaries based on reward signals	[145,146]
Attention Mechanism	Focus on important sentences for summarization	[74,75, 159]
Transformer-based Models	Utilize transformer architectures for improved summarization	[149,156]

4.6. Human-in-the-loop optimization for high-stakes construction NLP tasks

With increasing application of Construction NLP for Contract Analytics; Hazard Detection; and Dispute Resolution; there is a rapidly increasing need for Human-in-the-Loop Optimization Frameworks. RLHF (Reinforcement Learning with Human Feedback) presents a robust method to optimize Large Language Models (LLM) through the inclusion of expert feedback in the Training Loop [61,164]. The goal of this method is to ensure LLMs not only learn linguistic patterns, but also produce outputs consistent with domain specific reasoning an absolute necessity in high risk scenarios like Contract Dispute Resolution or Safety Incident Detection [83].

As an example, the use of RLHF to enhance the accuracy of classifications (i.e., accurate classification) for ambiguous clauses in construction contracts or to better identify latent risk indicators in daily reports improves the human-AI interaction; as well as, increases transparency, explainability, and trust in the automated decision system particularly, in legally sensitive or safety-sensitive construction projects. Additionally, with increasing availability of open source RLHF toolkits and expert feedback mechanisms, incorporating RLHF into the workflow of construction NLP represents an attractive method for improving the reliability of model performance [42,45] (refer Table 7).

Beyond improving model output quality, RLHF also addresses some of the practical challenges in dataset scarcity and generalizability two key limitations often encountered in construction NLP. By incorporating real-time expert corrections and preferences during model fine-tuning, RLHF reduces the dependency on large annotated corpora, which are often unavailable or confidential in construction domains [43,167]. This dynamic feedback loop allows for more agile model development, where systems can be updated based on evolving regulatory standards,

Table 7
Techniques and Purposes in Human-in-the-Loop NLP for Construction.

Method	Purpose	Reference
Reinforcement Learning with Human Feedback (RLHF)	Align model outputs with expert preferences in high-risk tasks	[61,164]
Expert-in-the-Loop Fine-Tuning	Continuously adapt NLP systems to domain-specific language and context	[165,166]
Human-Guided Error Correction	Improve model reliability through iterative expert feedback on outputs	[43,44,61]
Interactive Feedback Mechanisms	Enable responsive updates to model behavior in evolving construction scenarios	[44,164, 165]
Evaluation via Human Preference Scoring	Assess quality of model predictions beyond automated metrics	[42,82, 167]

site-specific terminology, or project-specific reporting styles. As construction firms increasingly adopt AI-driven solutions, embedding human-in-the-loop frameworks ensures models remain adaptable, context-sensitive, and aligned with stakeholder expectations [83,164].

4.7. Comparison of NLP approaches in construction engineering

Table 8 presents an overall summary of all the most up-to-date developments in NLP-driven research within the area of construction engineering by providing a comparative table of all the most recent techniques for NLP, datasets used for testing, evaluation criteria, and the applications in which they are being employed. The use of word embedding techniques and traditional machine learning models has been replaced with the advent of deep neural networks and the development of transformer-based models that provide better contextual understanding [12,51]. Additionally, the use of multiple types of datasets in each study is demonstrated (incident reports/maintenance logs/contracts/RFI's/BIM related textual documents) and the variety of methods used to evaluate performance (F1 score/Precision – Recall/-ROUGE/Cosine Similarity/Human Expert) demonstrate the range of NLP techniques that can be utilized to perform a wide variety of construction activities (Safety Analysis/Contract Interpretation/Linkage between Design Information/Compliance Checking/Automatic Summarization) [168–172]. The data presented in this table will allow researchers to see the current state of the art and also identify potential gaps and opportunities for developing standardized datasets and benchmarking frameworks for specific NLP applications in the construction industry [173–176].

Table 8
Recent NLP methods, datasets, evaluation metrics, and application domains in construction engineering.

NLP Method / Model	Dataset Used	Evaluation Metrics	Application Domain in Construction
BERT, RoBERTa, DistilBERT	Safety incident reports, OSHA datasets, proprietary site logs	F1-score, Precision, Recall, Accuracy	Safety hazard detection, root-cause classification, injury prediction
Transformer Models (T5, GPT-based, XLNet)	Construction specifications, contracts, RFIs, BIM textual data	BLEU, ROUGE-L, perplexity, human expert evaluation	Contract clause understanding, automated compliance checking, specification summarization
Bi-LSTM / LSTM / GRU Networks	Daily site reports, defect logs, progress reports	Accuracy, AUC-ROC, Cross-entropy loss	Progress monitoring, defect classification, project performance prediction
Word2Vec / GloVe Embeddings + Traditional ML (SVM, RF, NB)	Accident narratives, maintenance logs, facility management text	Precision, Recall, Macro-F1	Safety text classification, equipment failure prediction, FM documentation analysis
Named Entity Recognition (spaCy / BERT-NER)	Contracts, change orders, dispute case documents	Entity-level F1, Exact Match Score	Entity extraction for contractual parties, obligations, material specifications
Topic Modeling (LDA, NMF, BERTopic)	Scopus/WoS publication datasets, organizational reports	Topic coherence (Cv), UMass score, diversity metrics	Research trend mapping, knowledge discovery, risk trend identification
Dependency Parsing / Rule-based NLP	Codes & standards (IBC, NEC), BIM object metadata	Extraction accuracy, rule-matching accuracy	Automated code checking, design constraint validation
Semantic Similarity Models (SBERT, USE)	RFI–specification pairs, design documents	Cosine similarity, semantic accuracy	RFI response automation, design–spec linkage, information retrieval
Summarization Models (PEGASUS, T5)	Long construction claims, H&S reports	ROUGE-1/2/L, human evaluation	Claim summarization, safety briefing generation
Text-based Multimodal Models (Vision-Language: CLIP, BLIP)	Images + textual safety notes, inspection reports	Recall@K, multimodal matching accuracy	Safety violation detection, visual-text compliance checking

5. Future research trends

The field of NLP in construction engineering has seen substantial advancements in recent years. These advancements have used AI-powered text analysis approaches to improve several facets of building projects. Nevertheless, with the ongoing expansion and development of the subject, there exist distinct study domains that provide distinctive difficulties and prospects for further investigation [177–179]. This section provides a more comprehensive analysis of these domains, emphasizing the need for further investigation to effectively respond to the changing requirements of the construction sector. This Table 8 presents a complete summary of the contributions made by previous research in the field of NLP in the context of building. It also presents the findings of this study which served to fulfill the research gap. This also functions as a significant point of reference for forthcoming research aspects within the domain of NLP as it pertains to the application of construction engineering Table 9.

Construction projects produce a substantial volume of data, including textual data, photographs, sensor data, and other forms of information. Future research should prioritise the development of sophisticated methodologies aimed at effectively integrating and comprehending data derived from a wide range of modalities [68,71,180]. The integration of many components is crucial in order to get comprehensive insights into building projects. An enhanced comprehension of project progress, quality control, and safety compliance may be achieved by integrating textual data extracted from project documentation with visual media such as photographs and videos captured at construction sites [181–183].

In the business of construction, transparency and interpretability are extremely important as some of the decisions in the area are highly significant, highly influential to aspects such as safety, cost and the success of a project in general [184–186]. While future research needs to prioritize the development of such NLP models that will not only be precise in its forecasts but also will explain the rationale for its decision making in order to advance the field [55,57,58]. In the domain of NLP, Explainable Artificial Intelligence (XAI) can be deployed to fix the problem of complicated AI technology and understandable right results in the construction companies [187–189]. An example is that for the task of risk assessment, an AI model must be able to explain how it identified a particular danger [66,69,70]. By doing this this would allow the experts in the construction industry to make judgements based on the model underlying thinking. For building NLP, the realisation of XAI requires pioneering studies on the interpretability of model, explanations about decisions and creating of user-friendly interfaces.

The construction projects have dynamic nature that that they are characterized by rapidly changing circumstances within the on-site environment. Given as such, future research pursuits to overcome this issue have to focus on real time applications in NLP which provide instant insights and decision-making support. These apps provide the ability of consistent analysis and processing of textual data that experts in the construction industry can utilize so as to quickly address emerging

Table 9
Comparative contribution of research.

Aspect of Contribution	Existing Studies in NLP in Construction	Findings from this Study
Multimodal Integration	There is a lack of sufficient focus on the efficient integration of multiple data modalities.	Emphasised the need of using sophisticated methodologies to effectively merge textual data with visuals and sensor data in order to get comprehensive insights.
Explainable AI	Limited discourse on the advancement of NLP models with regard to their interpretability inside the construction domain.	This statement underscores the significance of Explainable AI (XAI) within the realm of building NLP, as it pertains to the development of AI models that are transparent and interpretable.
Real-time Applications	There is a limited body of research that investigates the use of real-time NLP systems in the context of agile decision-making within the construction industry.	The author advocates for the use of real-time NLP applications in order to provide timely insights and assistance for building projects that are subject to constant change.
Customized NLP Models	This study primarily focuses on the development of specialised NLP models that are especially designed to cater to the unique terminology and contextual requirements of the construction industry.	This emphasises the need of developing domain-specific NLP models that are trained using construction-related textual data.
Data Interoperability	Insufficient focus on effectively resolving data interoperability obstacles within the construction sector.	The need of addressing data interoperability challenges is underscored in order to facilitate the smooth integration of textual data originating from various sources.
Domain-specific Lexicons	Limited investigation of specialised lexicons pertaining to construction terminology and expressions.	It is recommended that the creation of domain-specific lexicons be considered as a means to improve the precision of NLP applications within the construction industry.
Sentiment Analysis	The use of sentiment analysis for the purpose of assessing stakeholder attitudes and emotions in building projects is constrained.	The proposition was made to endorse the use of sentiment analysis as a means to enhance stakeholder involvement and optimise project results.
Knowledge Graphs	Limited discourse exists on the development of knowledge graphs for the purpose of representing and organising construction-related information.	This study emphasises the potential of knowledge graphs in organising and retrieving construction information from textual sources.
Privacy and Ethics	There is a limited body of research that specifically examines the privacy and ethical implications associated with NLP in the construction industry.	The need for ethical protocols and privacy protections in NLP applications within the construction industry is underscored, particularly in instances involving sensitive data.
Human-AI Collaboration	There is a dearth of research examining the many types of cooperation between humans and AI specifically within the context of building jobs.	The user expressed support for doing research on the efficacy of cooperation between construction experts and AI systems in order to improve decision-making processes and boost overall efficiency.

problems and leverage potential opportunities [56,59,65]. The lexicon of the construction sector is unique and it has its own set of problems similar to any other sector. To get improved accuracy and relevance of these models, it is suggested that the training of these models on construction related text data such as project reports, specifications and industry publications would be recommended [80,190,191].

Data interoperability is a notable challenge in the implementation of NLP within the construction industry [27,192,193]. Construction projects include a diverse range of data derived from several sources and forms, such as Portable Document Format (PDF) files, photographs, spreadsheets, and other relevant mediums. Future research should prioritise the development of standards, protocols, and methodologies to ensure the smooth integration and analysis of textual data from various sources. The resolution of data interoperability obstacles will facilitate the development of more efficient and effective NLP systems within the field of construction engineering. Further, following are the most critical future trends of NLP in construction according to the outcomes of literature analysis.

5.1. Integration of deep learning models

Integrated deep learning models have the potential to greatly advance the area of construction engineering via the seamless integration of NLP with sophisticated deep learning methods. Integrated Deep Learning Models play a crucial role in many construction applications, necessitating an exploration of strategies for their development using NLP:

- **Document Management:** It is a crucial aspect of construction projects since they create substantial volumes of textual data, including project plans, specifications, and contractual papers. Integrated deep learning models have the potential to be used for the automated classification, extraction, and summarization of this information. This system enables streamlined document management and retrieval processes, hence alleviating the administrative workload placed on project teams [102].
- **Risk Assessment:** In the world of building it is the full evaluation of quite a few elements. Deep learning models can be integrated to do a variety of data textual analysis such as risk reports, contract papers and environmental data. New options for enhancing these models exist by incorporating real time project updates into the analysis process. Using this integrated analyses, it is possible to have a more complete identification of likely hazards and to take proactive steps to mitigate the risk [194–196].
- **Real-Time Decision Support:** Because of changing circumstances, quick decision making is often necessary for RTM in construction projects. Deep learning models integrated with the real time textual and sensor data, and helps to process a wide variety of data like weather updates or condition of equipment's. It enables project managers to make optimal decisions quickly enough so that the occurrences of delays and interruptions are averted [8,107].
- **BIM Integration:** By integrating BIM and Deep Learning Models, BIM contributes to the linkage between NLP analysis of textual input and information obtained from Building Information Models. Again, the implications for an improved comprehension of the subject matter result in the integration of construction plans and progress that benefit from their improved coordination and fewer mistakes [12,107].

Methodologies such as transformer based architectures and deep learning frameworks such as TensorFlow and PyTorch are included [12,197]. The natural language comprehensions and contextual extraction from textual input of the proposed models shall be made possible by sophisticated NLP components. Besides, photos and videos will be analysed and interpreted using computer vision modules to improve the overall comprehension of building projects [14,198]. While the integration of data from a number of sources, combined with other real-time information, has the ability to significantly raise effectiveness, accuracy, and cooperation in the field of construction. Better project results and reduced expenses may be sourced from it.

Continuous progress in deep learning architectures such as transformers in various NLP tasks such as text classification, named entity recognition, sentiment analysis, and machine translation, the task of

question answering seems to be relatively less explored [199–201]. A lot of the scholarly works have been studied in the integration of deep learning models (for example, transformers) in NLP applications in the field of construction engineering [101,202,203]. It placed a transformer based model introduced for the construction task of document classification [50]. Their research findings found that transformer architectures, such as BERT (Bidirectional Encoder Representations from Transformers), increased the precision of the document classification on the basis of their ability to absorb complex linguistic relations and semantic dependencies of construction project documents.

Similarly, A study employed the use of transformers to automate the extraction of pertinent data from construction project documents [196]. Transformers have been included and which have considerably improved precision and effectiveness of the information extraction in order to aid in informed decision making as well as improving the project management capabilities. The aforementioned benefit has been utilized in diverse construction engineering NLP tasks. As an illustration, N. Xu et al. [189] utilized transformers to conduct sentiment analysis on stakeholder communication within the context of construction projects. To tackle these challenges, it will be necessary to make progress in model optimization techniques, implement efficient training procedures, and create lightweight models that are appropriate for use in construction environments with limited resources [127,204].

5.2. Multimodal analysis

The discipline of construction engineering may greatly benefit from the use of multimodal analysis, which involves the integration of many data modalities including text, photos, and videos. In this regard, the positive aspect lies in the fact that it provides a means to fully understand synchronized building projects that are very complex. Sometimes it is necessary to use the Multimodal Analysis method, and in that case, one must use the NLP methods for creating models:

- **Progress Monitoring:** Progress monitoring in construction projects include a wide range of data sources, such as project paperwork, visual records of work locations, and other reports [205,206]. Multimodal Analysis allows us to associate textual information, for example, project status reports, with pictures or movies (visual elements) taken at the building site [3,5]. This assists project managers in gaining an overall sight of project progress and swiftly locating possible troubles.
- **Defect Detection:** The identification of defects in construction materials or workmanship is critical to assuring the overall quality of an undertaking. Multimodal inspection report analysis can include both photographic/video-based documentation, as well as textual documentation, to enable the accurate identification/classification of issues [7] and by providing a visual context for the issue, it provides a basis for identifying defects, and expedites remedy.
- **Safety Compliance:** Ensuring that the construction industry adheres to all appropriate safety protocols is critical. Multimodal analysis allows for the study of safety documentation along with video information collected by cameras at a site (i.e., on-site cameras) to verify compliance with safety procedures [46,54]. Identifying potential safety hazards ahead of time and taking corrective action can prevent incidents from occurring.

NLP approaches will assume a pivotal role in the development of Multimodal Analysis models in next research endeavours. The use of NLU techniques will be employed to extract valuable insights from textual material, while Computer Vision algorithms will be utilised to analyse and interpret photos and videos [26,207]. The integration of advanced deep learning models, like as Transformers, would allow the amalgamation of data from several modalities, hence enabling a full examination of building projects [31,35]. The use of integrated models in the construction sector is expected to provide a comprehensive comprehension of construction processes, hence facilitating superior decision-making, heightened quality control, and enhanced project management.

Moreover, it will move closer to the integration of multimodal analysis in NLP applications for the construction engineering field. Wide class of data is created in construction project, including textual information, visual representations (images, videos), and sensor data. Consequently, along with computer vision and audio analysis, NLP methodologies are used to enhance construction industry experts' comprehension of project related data in holistic manner. There have been many scholarly investigations conducted to investigate the incorporation of multimodal analysis into NLP applications in construction engineering area. Z. Z. In Chen et al. [22] they developed a multimodal methodology to evaluate construction risks by text analysis, image and video analysis.

In a similar vein, Fan et al. [27] employed a multimodal methodology to monitor construction progress. The researchers in this study employed a combination of NLP techniques and image analysis to examine textual project reports and visual images of construction sites [51,208]. The construction industry professionals will be able to understand the project progress visually from the images of the construction site or 3D model, note potential hazards and estimate the amount of work done at the construction site [5,81]. Additionally, it enables the improvement of accuracy and relevance of NLP applications for construction engineering [24,106]. One of such is the development of a multimodal sentiment analysis framework for stakeholder communication in construction projects by Bikaun et al. (2022) [21]. In their study, the researchers were able to combine text analysis and sentiment analysis of social media images to portray the emotional and sentiment messages by different stakeholders. Because all these types of heterogeneous data need to be integrated and processed, specialized methodologies are required to fuse data extract features and integrate the resulting structures [56,69,111]. Future work should be focused on the creation of effective and scalable means for the management of multimodal data within the scope of NLP applications in the domain of construction engineering.

But the challenge in order to make sense out of the provided information from different modalities and to derive useful insights is quite tough. Future work will explore how information could be integrated from diverse modalities in an integrated, human–intelligible way. To overcome these challenges, it will be necessary to make progress in computational methodologies, parallel computing systems, and optimized algorithms for multimodal analysis. Future research in the field of integrating multimodal analysis in construction engineering NLP applications should prioritize the resolution of these challenges and the exploration of potential solutions [35,209,210].

5.3. Explainable AI in NLP

The use of XAI presents significant opportunities for effectively tackling key difficulties in the domain of construction engineering. In some particular scenarios, the execution of this approach is not only advantageous but often necessary for the following reasons:

- **Risk Assessment and Safety:** The prioritisation of safety is of utmost importance in the field of building. Therefore, XAI would be an effective tool to provide overarching explanation to risk assessment models, and to help stakeholders understand why certain hazards are identified from the various risk assessment models [49,56]. The provision of openness is of utmost importance in facilitating well-informed decision-making and bolstering safety standards.
- **Project Delays and Cost Overruns:** Construction projects are well recognised for their propensity to experience delays and cost overruns. XAI can generate prediction model with elucidations on possible delays like adverse weather condition or interruption in the supply chain [25,26]. Gaining an understanding of the underlying

causes of delays facilitates the implementation of proactive strategies.

- **Resource Allocation:** The distribution of resources plays a crucial role in ensuring the efficiency and success of a project. XAI can provide interpretable suggestions on resource allocation, that is, how many equipment to deploy or where to allocate labour [25,63].
- **Quality Control:** XAI has potential to improve quality control processes by enabling the description of the underlying standards that are used to assess the quality of building materials and workmanship [40]. The use of this transparency measure guarantees the continual adherence to quality standards.

Models applicable to the construction field would be facilitated by the utilization of several approaches such as rule based systems, model agnostic approaches and enhancement for visualisation tools. The development of the transparent decision rules are facilitated by rule based systems. On the other hand, model-agnostic approaches such as LIME (Local Interpretable Model-agnostic Explanations) may be used to provide explanations for the predictions made by intricate machine learning models [37,38]. Moreover, the use of sophisticated visualisation tools enables stakeholders to actively engage with model results and get insights into the variables that impact a certain choice.

Recent studies have extended these principles by applying explainable NLP models to regulatory review and permit approval processes. By using frameworks such as SHAP and LIME, these models provide transparency in decisions related to clause detection, risk flagging, or zoning compliance interpretation [211]. When an NLP model recommends rejecting a permit due to missing environmental clauses, the system highlights the exact sentence or section driving that conclusion. This not only builds user trust but also aids regulators in validating the AI's reasoning and accelerating the review pipeline [212].

Due to the rise in prevalence of AI-based NLP models, explainable AI in the context of NLP applications in the area of construction engineering has attracted significant interest of the market. Explainable AI is the field aimed to increase transparency and comprehensibility of AI models' decision making processes to their human users. Baek et al. [65] introduced the LIME (Local Interpretable Model-agnostic Explanations) framework, which offers explanations for individual predictions made by NLP models.

In a similar vein, the SHAP (SHapley Additive exPlanations) framework was introduced by Park et al. [72] as a means to offer game-theoretic explanations for NLP models. The researchers' study showcased the utilization of SHAP in the domain of text classification and sentiment analysis tasks. The framework assigns significance scores to input features, thereby offering insights into the contribution of each feature to the decision-making process of the model [86,87,213]. Recurrent neural networks (RNNs) and transformers have shown very promising performances on a large number of NLP tasks, and thus have become deep learning paradigms for NLP. Scholars have then suggested various methodologies to handle this specific barrier. Guo et al. [214] proposed the utilization of an approach known as "attention" within the framework of sequence-to-sequence models for machine translation. Attention-based methodologies have been expanded to encompass additional NLP endeavors, including sentiment analysis and named entity recognition, with the aim of offering justifications for the predictions made by the model.

In addition to techniques specific to each model, there have been proposed post-hoc explainability methods aimed at offering explanations for NLP models. M. A. A. S. Ali [215] proposed the notion of "anchors," which refer to interpretable and minimal input subsets that possess the ability to predict a specific class. The integration of explainable AI has the potential to augment the acceptance and implementation of NLP models within the domain of construction engineering. However, methods that offer a compromise between the comprehensibility of results and the richness of models can yield enlightening insights with a fair amount of precision in the domain of NLP [84,216,217]. Furthermore, the advancement of domain-specific explainable AI techniques specifically designed for the distinct attributes and demands of construction engineering can augment the efficiency and pertinence of the explanations offered by NLP models [25,63].

5.4. Domain-specific NLP models

NLP domain specific models are developed to specifically meet and address the different problems and specialized needs in the construction area. There are several domain specific NLP models which have a specific place in the construction area and can solve construction related problems. The development of DSMs for the construction area is also influenced by various specific application areas and the respective requirements. Therefore, DSMs will be created for each specific domain. There are several ways of creating these models:

- **Construction Terminology:** The Construction Industry has a large number of specialized terms and definitions in the realm of construction. Specific terminology in the construction industry is required for the production of DSMs to provide the capability of creating automatic documentation and reports [24,34].
- **Regulatory compliance:** The use of domain specific models in this field of construction is very important as there are many regulations and requirements for the work that they do; however, models can be used to interpret complex legal documents that help with compliance of all applicable laws domestically and internationally on all project levels [47,196]. By doing so this practice not only protects the companies from potential legal liabilities, but also will aid in speeding up the time needed to obtain a building permit.
- **Materials and Specifications:** Accurate and detailed information is required in order for the successful completion of construction projects, which relates to both materials and project needs. DSMs can be developed so that they may extract and understand a variety of types of data, such as technical specifications, material catalogues, and architectural blueprints [11,218], this enables precise procurement of materials and project planning.
- **Equipment and Machinery:** Construction companies are heavily reliant upon heavy machinery and equipment to successfully operate. Because domain specific models can be trained with a variety of information; examples include equipment manuals, maintenance schedules, operating data, etc., [27,33]. they have the potential to improve the efficiency of equipment usage, while also enhancing safety at construction sites.

In upcoming advancements in the area of Domain Specific Models for building, there are two ways to build: using fine-tuning and/or transfer learning techniques to train a pre-trained language model. These same models will then undergo domain adaptation so they can develop their own knowledge of construction-specific terminology, construction-specific rules and construction-specific documentation [50,55]. Successful training of these models will require a collaboration between both domain experts and the various stakeholders from the building industry [155,219]. The resulting NLP models will be specifically adapted to the complexity of the building process. This will lead to enhanced project efficiency, adherence to regulations, and accuracy in an area where specialised expertise holds utmost importance.

Further the emergence of domain-specific NLP models, which are trained on datasets specifically tailored to the construction industry, has garnered significant interest as a prospective direction in the realm of construction engineering [156]. Tanniru et al. [190] proposed the development of a domain-specific NLP model designed specifically for the classification of construction project documents. We find that a construction specific NLP model is more successful than general purpose NLP models, for the most part because it is able to capture the building specific language and terminology [2,90,220].

The NLP model designed for a specific domain demonstrated superior accuracy and relevance in extracting construction-specific information when compared to NLP approaches that are designed for general purposes [73,221]. This is further reinforced by the work of Kessler and Béchet (2025), who developed a deep learning framework for automatically extracting hypernym relations to construct domain-specific glossaries. Their model efficiently captures hierarchical term structures and contextual meanings within engineering texts, improving NLP model precision in classification and semantic understanding tasks [222]. This approach enhances the relevance of NLP applications in construction settings where linguistic variability, technical jargon, and context-dependency present consistent barriers to general-purpose NLP tools. It should be the future research direction to develop and curation of large and reliable construction specific datasets covering a wide range of NLP tasks, including many aspects of construction engineering and according to these NLP tasks [134,189]. All these resources cover both the linguistic and the contextual sides of the construction domain, providing aid to domain specific NLP models in more comprehending and analyzing textual data of the construction domain [22,223,224]. An example of this can be seen in the work of Burgueño et al. [151], who introduced a word embedding model specifically designed for construction-related language, known as ConWord2Vec.

5.5. Real-time NLP applications

Real-time NLP models offer essential capabilities to analyze, monitor, and make decisions on real-time information in the Construction Sector. The following outlines the application of Real-Time NLP models in the Construction Industry; the necessity of implementing them; and the methods used to develop them:

- **Safety Monitoring:** For safety monitoring, safety incident reports, on site communications, and other sensor data from construction sites are analyzed as they happen with real time NLP models [21, 225]. These models contribute to the improvement of construction site safety by promptly identifying and reporting safety problems, hence enabling timely interventions in the event of accidents or potential risks.
- **Project Progress Tracking:** I use real-time NLP models to analyse textual project update to monitor project development. The sensor data is then integrated with these updates to allow the immediate generation of insights of the advancement of the project [21,28]. It is essential for project managers and stakeholders to make timely choices, adapt schedules, and guarantee projects remain on course.
- **Quality Control:** It plays a very important role in construction industry in which identification of quality faults instantly is a foremost factor. Real-time NLP models can be able to read reports and pictures generated on-site and detect errors right away, and fix them as well [23,29]. This capability ensures the maintenance of project quality and the reduction of rework.
- **Environmental Impact Assessment:** Construction projects are required to comply with environmental rules as part of the Environmental Impact Assessment process. Such real time NLP models are capable of actively watching environmental impact reports and sensor data and trigger warnings in real time about the detrimental environmental impacts and actively suggest measures that minimize them [70].

In forthcoming times, the advancement of Real-Time NLP models for the construction industry will need the integration of streaming data processing methodologies with real-time analytics [60,74]. For that purpose, we will use cutting edge technologies, Apache Kafka for the stream processing and advanced NLP frameworks to speed up the analysis of textual data. Moreover, IoT devices and sensors would be included into building sites to provide consistent flow of data to be real time NLP analyzed [80,215]. The use of these models would provide proactive decision-making and prompt solutions to developing difficulties, therefore enhancing the efficiency, safety, and quality of building projects, while concurrently mitigating risks and minimising delays. Supporting this need, a study presented a real-time NLP system capable of scanning communication logs and sensor data to proactively detect emerging construction risks. Their implementation successfully integrated into project dashboards, offering real-time alerts to site managers about safety violations, schedule delays, or material shortages [226].

These trends are also supported by other research that suggests using sensor logs in real time and integrating them into NLP models to support both operationally and safe work environments. They do this by comparing text-based logs of equipment shifts, tool use records, and incident reports, with real time sensor output data to identify inconsistencies and/or anomalies in the data. For example, if it is recorded in the log that a piece of equipment was "turned back in," however, that same equipment is still active according to the sensor data then the system will flag an error or potential safety issue. Using multiple sources for validation creates better risk identification, increases accountability, and improves incident prevention methods for an operating jobsite [70].

Construction engineering has experienced an increasing trend of using real-time NLP applications [65,74]. These applications allow users to process and analyze text-based data in real-time from many different sources including but not limited to; social media posts, project communication logs, and sensor readings. Many scholars have investigated the utilization of real-time NLP applications in the area of construction engineering, and although many of the real-time NLP applications have been studied under subtopics of risk management, stakeholder engagement and project controls, Bhatia & Pinto (2021) [60], developed a real-time NLP application, which proactively identifies and provides information regarding risks and the underlying cause of those risks so that construction professionals can take prompt action to mitigate those risks.

Like Chen & Tsai's [163], in addition to improving stakeholder engagement for construction projects by developing an NLP method that can be utilized in real time, they used both Sentiment Analysis and Topic Modeling to analyze the social media feeds and real-time communication logs in their study. In the same way, through the use of real-time processing and analysis of textual data, professionals in the construction industry will have the ability to identify emerging issues, trends, and patterns as well as other types of information. Another problem is how to integrate real-time NLP applications with all of the different technology and systems currently being used in the construction industry. Construction projects can include many different forms of data such as BIM data, IoT sensor data, and project management systems; therefore, it would be beneficial to determine which form(s) of data could be most easily integrated with real-time NLP applications [64].

Moreover, the reliability and speed of the results from real time NLP applications will contribute to quick and accurate decision making. Due to the nature of the processing and analysis being done in "real time," there are strict deadlines as to when an application needs to produce results. As such, one of the largest challenges that exist for ensuring high levels of accuracy within all real time NLP applications is that it has to be achieved in real time. Therefore, future developments in real time NLP applications in the construction engineering industry, should consider multiple areas. Improvements in the development of streaming NLP algorithms and methods can also lead to improvements in how efficiently and effectively large amounts of data can be processed and analyzed in real time. Moreover, advancements in real-time dashboards, alerts and interactive visualizations, can also contribute to better decision making at the time of need for construction engineering projects and enhance collaboration between all parties involved with the project [28,227,228].

5.6. Human-in-the-loop NLP: the role of RLHF in construction

The integration of large language models (LLMs) into construction

industry workflows has unlocked a new era of intelligent document interpretation, automated data extraction, and enhanced communication [43]. However, this advancement also brings forth a critical challenge: how to ensure that these AI models behave in a manner that aligns with the nuanced requirements, safety standards, and regulatory expectations of the construction domain. One promising solution to address this alignment problem is Reinforcement Learning with Human Feedback (RLHF) [44].

In the construction sector, RLHF can be highly impactful in areas such as:

- Contract Analysis and Legal Interpretation: Construction contracts often contain dense, technical, and legally sensitive language. RLHF allows models to better understand and align with human interpretations of ambiguous clauses, liability structures, or compliance mandates [83].
- Safety Documentation and Risk Mitigation: Safety manuals, incident reports, and hazard assessments require cautious and context-aware interpretation. Human-in-the-loop feedback can train models to avoid hallucinated outputs and prioritize critical safety cues [164].
- Specification Compliance and Technical Reviews: Models can be guided to highlight discrepancies between design documents and project specifications, with domain experts reinforcing correct interpretations and penalizing misleading ones [43].

For construction NLP, this opens future research avenues such as:

- Designing construction-specific reward models for RLHF.
- Developing interactive annotation platforms that allow engineers and planners to fine-tune outputs during workflow integration.
- Creating benchmarking datasets that compare the performance of RLHF-augmented LLMs against traditional NLP models in real-world construction scenarios [43,82].

Furthermore, RLHF can play a pivotal role in enhancing the interpretability and scalability of AI solutions in high-stakes construction scenarios. For instance, in contract dispute resolution, RLHF can ensure that LLMs align with legal reasoning and contractual precedents, reducing ambiguity in automated clause interpretation. In real-time hazard detection, integrating expert feedback can help NLP systems adapt dynamically to site-specific language and context, making safety assessments more responsive and actionable. These use cases highlight the importance of human-AI collaboration frameworks, with RLHF offering a path toward trustworthy and context-aware deployment of NLP in construction.

In conclusion, RLHF holds transformative potential for construction NLP by embedding expert reasoning directly into model behavior. Its application can significantly reduce operational risks, increase legal and technical compliance, and foster stakeholder confidence in AI-assisted construction management.

Construction Engineering has an important knowledge gap with regard to the lack of common (standard) benchmark data sets in NLP. In addition to utilizing various types of internal safety reports, contracts, RFI's, progress logs, and defect narratives, many of the data sets used by researchers are proprietary, unstructured, and therefore cannot be utilized by other researchers to provide comparisons and reproducibility. Unlike General NLP Domains that have well established benchmarks like SQuAD, GLUE, and CoNLL; Construction does not have access to well defined, curated data sets for tasks such as Hazard Classification, Contract Clause Extraction, Automated Code Compliance Checking, and Design-Text Alignment. There are very few data sets available that are either partially accessible, or contain some type of BIM related textual corpus. These are insufficient to support the consistent benchmarking of modern transformer-based models or LLMs. This indicates a strong need to develop Domain Specific, Anonymized and Openly Available NLP Data Sets to Support Model Evaluation, Provide Fair Comparison Between Methods, and Advance Development of Robust Language Models Focused on Construction.

5.7. Integration with building information modeling (BIM)

BIM Integrated NLP models use the capabilities of NLP in conjunction with comprehensive data derived from BIM models, hence introducing transformative advancements in construction project management and decision-making processes. The following are some use examples for the integration of BIM and NLP models in the construction industry:

- **Design Validation:** It may be achieved via the use of BIM-integrated NLP models, which have the capability to assess design requirements in relation to BIM data and construction plans [22,36]. This guarantees that construction projects conform to the original plan and eliminates expensive inconsistencies between design intent and execution.
- **Change Management:** Change management is a common occurrence in construction projects, often necessitating modifications and adjustments [9,107]. NLP models in BIM Integrated possesses the capability to examine modification requests, compare to the BIM model, and supply real-time insights on possible ramification on the project timeline, cost and resources.
- **Subcontractor Coordination:** Coordination of subcontractors in the building industry are complex. The use of work order matching with BIM data may allow BIM Integrated NLP models to be used for coordinating subcontractor operations [1,9]. This process ensures that there is congruence with project milestones and requirements.
- **Quality Assurance:** In the field of construction, quality assurance is very much important as it guarantees a quality of project to entire. NLP models of BIM Integrated capture the ability to compare quality control reports and BIM data to ascertain whether building activities complied with design standards and at the same time compliance requirements [2,15].

However, methods like semantic mapping and graphes based analytic could be used in the future in order to further BIM integrated NLP models development. This means the BIM models that will be discussed in the aforementioned models will be required to understand the semantic properties of BIM models and establish mapping between the BIM models and text that is written in plain language. The aim is to improve communication with the stakeholders and make sure that the building project is as the design aims at. Due to the rich data structures generated by BIM, BIM Integrated NLP models are expected to offer many benefits to the construction field such as increased project management, reduced errors, higher quality, easy construction processes and so on. Therefore, these models are considered essential in the field.

5.8. Evaluation frameworks for large language models (LLM) in construction NLP

The use of Large Language Models (LLMs) in the construction industry expands upon the potential for NLP-based decision making. Although LLMs, like GPT, BERT, and LLaMA have shown strong generalization across many types of linguistic tasks, their value in construction-related fields must be tested through robust evaluation methodologies [184]. Due to the high level of specificity and the critical nature of the construction engineering environment, it is unlikely that traditional methods of evaluating these models on standard benchmarks would adequately represent the true performance of the models in these areas [167].

State-of-the-art benchmarks (such as MMLU, BIG-bench, HELM, and TruthfulQA) as identified by Laskar et al. (2024) give insight into the capabilities of LLMs in reasoning, factual consistency, and commonsense logic. However, most of these benchmarks do not contain

domain-specific tasks that are necessary for the AEC sector [165]. Most notably, the performance of LLMs in extracting project timelines; interpreting clauses from specifications documents; processing RFIs (Requests for Information); and identifying hazards from safety logs has been poorly studied. Therefore, researchers need to create construction-specific benchmarks that reflect real-world tasks, regulatory language, and engineering constraints [82].

Finally, another key factor in the evaluation of LLMs is providing context and understanding the interpretation of the model's output. Since construction involves collaboration among multiple stakeholders (i.e., engineers, contractors, government entities), models need to produce transparent explanations of why they produced a particular output. As such, incorporating explainability metrics such as model faithfulness (using techniques such as LIME or SHAP) should become a primary aspect of LLM evaluation in this area [45]. Additionally, evaluation frameworks must assess the model's capacity to perform well in various subfields of construction including structural design, MEP coordination, and environmental reporting [42].

6. Implications

There are several possible implications from this research regarding how the use of NLP for Construction Engineering could affect the Construction Industry. A potential implication is that using NLP-based automation to classify and categorize documents will provide an opportunity to increase productivity when it comes to processing large volumes of textual data associated with construction projects. An additional benefit of this approach is that it will reduce the amount of time to review textual data; therefore, reduce the chances of errors caused by humans. Structured documentation will allow for quicker access to critical information associated with a project allowing for better-informed decision making and ultimately improve the overall productivity of a project.

A construction organisation is able to monitor and control the feelings of their stakeholders, in real time, by the use of Sentiment Analysis which is one of the many applications of NLP. A positive sentiment has the potential to build satisfaction and trust within stakeholders whilst negative sentiments present an opportunity for a stakeholder to rapidly voice their concerns and potentially resolve them. In addition to the potential improvement in stakeholder involvement, a positive sentiment also presents opportunities to improve operational efficiency of projects. Information relevant to building projects can be automatically accessed using NLP techniques from a variety of documents and used to identify and assess risks associated with the project (magnitude of the hazard; probability of the hazard occurring; effects of the hazard). The outcome of this process is the development of improved proactive risk management strategies resulting in fewer interruptions to projects and therefore successful project completions.

Construction workers may be able to manage large volumes of textual information using NLP as a tool for providing them with a means to utilize data-driven analysis for making decisions about their work. Many of the NLP tools being used today are used primarily for extracting "key" information (i.e., names; numbers; etc.) from contract documents; specifications; and project reports. Data-driven decision-making will enhance the quality of project planning; resource allocation; and budgeting for construction projects. Additionally, document management systems utilizing NLP technology provide an opportunity to standardize and systematize methods for managing documents which will improve communication and collaboration among all stakeholders on a construction project and disseminate knowledge and best practices to enhance overall performance of construction projects.

7. Limitations

While this literature review has provided an extensive overview of NLP research in construction engineering, there are several potential limitations associated with it which provide important context to understand the implications of the review's findings. Firstly, the review is limited in terms of the extent to which published literature was accessible; many of the NLP studies related to construction use proprietary datasets (e.g., internal safety logs, contractual documentation, etc.) that are not publicly accessible. As such, it is possible that studies that are directly relevant to this review were omitted from the review due to either lack of access to the dataset or failure to publish. In addition, the lack of standardized benchmark datasets in the construction domain makes cross study comparisons difficult; therefore, the ability to compare the effectiveness of various NLP models or evaluate the overall methodological quality of the studies is significantly hindered.

A second limitation of this review is that there are a multitude of ways in which researchers report their results and utilize various types of metrics to evaluate them; as such, the ability to make direct comparisons between studies is challenging and all conclusions should be viewed with caution. Further, many of the earlier studies reviewed employed traditional machine learning approaches or rules-based pipelines, which likely do not represent recent advancements in deep learning and transformer architecture, and thus the review contains temporal biases. The continually evolving nature of large language models and multimodal models will likely lead to additional biases in terms of representing emerging techniques in the literature.

Notwithstanding the limitations described above, the review outlines a number of areas of potential future research. There is a significant need to develop domain specific, anonymized, and open benchmark datasets that address common construction related tasks such as: safety hazard identification/classification, contract clause extraction, specification-RFI alignment, and automated code compliance checking.

The aim of this review paper was to conduct a comprehensive analysis of the advancements and potential implications of NLP within the domain of construction engineering. Nevertheless, it is important to acknowledge distinct constraints and prospects for prospective enhancements:

- **Knowledge Cutoff Point:** The present research relies on the existing literature available as of the knowledge cutoff date in 2025. Although this research provides an overview of the field's progress up to the specified date, it may not include the latest developments that have appeared thereafter. Subsequent investigations should strive to consistently refresh the outcomes by incorporating the most recent advancements.
- **Scope of Focus:** The main focal point of this assessment is on the use of NLP applications, with a special focus on their application within the domain of construction engineering. Although efforts have been made to include a variety of perspectives, it is important to note that the scope of the research may not comprehensively encompass all possible applications or interdisciplinary connections. Subsequent inquiries may broaden their purview to examine the cross-disciplinary ramifications and pioneering implementations.
- **Temporal Analysis:** Although the study has included temporal analysis, its main focus is in the examination of trends and developments rather than the full exploration of historical aspects. Further investigation is warranted to explore the historical progression of NLP technologies in the domain of construction engineering, in order to get a thorough comprehension of its temporal advancement.

The development of NLP for the construction industry can be significantly accelerated by promoting an open science practice for construction-based NLP research. Therefore, future studies should emphasize releasing anonymized data sets and curation of text corpora and benchmarks through platforms like GitHub, Zenodo, or institutional data repositories to increase the transparency and reproducibility of research results across different studies. In addition, providing code implementations, pre-processing scripts, model checkpoints, and

experimental configurations could allow researchers to reproduce results, evaluate methodology and improve existing research more rapidly. Open science approaches will be necessary to support a collaborative research environment among the construction industry and academia to achieve a common standard for evaluating research methods, reduce duplicate efforts and promote innovation in applying NLP to the built environment.

8. Conclusion

The examination of NLP implementations in the field of construction engineering demonstrates notable progress, emerging patterns, and prospective avenues for future exploration. NLP has emerged as a highly valuable tool within the construction industry, facilitating the extraction, analysis, and comprehension of textual data to provide support for diverse facets of construction projects. During the course of the analysis, a number of significant findings and patterns have surfaced, underscoring the significance of NLP in the field of construction engineering. To begin with, the progress made in NLP methodologies, including the utilization of deep learning models and transformers, has resulted in substantial enhancements in the precision and efficacy of NLP applications. The aforementioned models possess the ability to comprehend intricate relationships between sentences, effectively process substantial amounts of data, and produce outputs that are more attuned to the surrounding context. The utilization of sophisticated deep learning architectures, such as transformers, has exhibited encouraging outcomes in construction-oriented assignments, encompassing document categorization, sentiment assessment, and data retrieval.

In addition to a variety of other major advantages in construction engineering, NLP can provide document management, risk assessment, stakeholder engagement, decision-making and collaboration. As a result of these various processes it will be possible to derive many beneficial insights, as well as, automate several of the processes involved in construction projects, resulting in improved efficiencies, reduced project risks, and ultimately better outcomes for all construction projects. Below are three NLP-based methods that may provide a number of significant impacts in construction engineering including, automated classification/categorization of construction project documents; sentiment analysis of communication by stakeholders in construction projects; extraction of valuable information from construction project documents; natural language understanding for construction risk management; and, text summarization for documentation of construction projects.

Future NLP implementations in the construction industry must prioritize human-aligned learning strategies. Techniques like RLHF offer a structured path to integrate domain expert feedback into the training pipeline, ensuring that NLP tools are both accountable and robust in interpreting complex construction data.

(1) Enhanced Document Management: Automated document management, driven by NLP, effectively reduces the amount of time and effort required for manual sorting and categorization of documents. This practice not only enhances operational effectiveness but also mitigates the potential for human fallibility in the realm of document management, thereby guaranteeing the precise categorization and effortless retrieval of crucial project data. In addition, the utilization of NLP in document management systems facilitates enhanced collaboration among various stakeholders involved in a project. NLP facilitates efficient sharing and retrieval of documents within a team by offering a standardized and structured method for organizing documents. NLP-powered document management facilitates the accessibility of essential information, such as project specifications, contract terms, and reports, thereby enabling more informed and timely decision-making.

(2) Improved Stakeholder Engagement: NLP-based sentiment analysis allows construction organizations to discern the positive or negative sentiments conveyed by stakeholders. The process of identifying stakeholders is of utmost importance in comprehending the perceptions and emotions that are associated with the project. Construction organizations can cultivate an atmosphere of trust, collaboration, and satisfaction by proactively addressing the concerns of stakeholders, improving communication strategies, and nurturing positive relationships. This enhances the efficiency of project implementation, garners greater backing from stakeholders, and ultimately leads to the attainment of project goals and objectives.

(3) Facilitated Risk Management: NLP techniques facilitate the extraction of crucial information from various construction project documents, including plans, specifications, contracts, and reports. This function facilitates the automated process of identifying and analyzing potential risks, including their underlying causes, as well as developing strategies to mitigate them. Construction professionals can effectively manage risks by proactively identifying and addressing potential hazards. The implementation of facilitated risk management practices plays a crucial role in attaining project objectives, improving stakeholder satisfaction, and bolstering the reputation and sustained prosperity of construction enterprises.

CRediT authorship contribution statement

Ahsan Waqar: Writing – review & editing, Writing – original draft, Visualization, Validation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

This is a review article. All data analyzed in this study are already presented within the paper.

References

[1] K. Demertzis, S. Demertzis, L. Iliadis, A selective survey review of computational intelligence applications in the primary subdomains of civil engineering specializations, Appl. Sci. 13 (2023) 3380, https://doi.org/10.3390/app13063380.
[2] L. Jacques de Sousa, J. Poças Martins, L. Sanhudo, Portuguese public procurement data for construction (2015–2022), Data Br. 48 (2023) 109063, https://doi.org/10.1016/j.dib.2023.109063.
[3] L. Jacques de Sousa, J. Poças Martins, J. Santos Baptista, L. SanhudoTowards the Development of a Budget Categorisation Machine Learning Tool: A Review Porto, 4200-465, Portugal Springer Science and Business Media Deutschland GmbH, Faculdade de Engenharia da Universidade do Porto Lect. Notes Civ. Eng. G.C. A. A. M.C. P.J.S.N. P.P. P.2023, 10111010.1007/978-3-031-20241-4_8.
[4] S. Chowdhury, A. Alzarrad, Applications of Text Mining in the Transportation Infrastructure Sector: A Review, Information 14 (2023) 201, https://doi.org/10.3390/info14040201.
[5] F. Campos-Medina, I. Ojeda-Pereira, J. Quiroz, J. Guzmán, International trends in mining tailings research through machine learning method: retrospective or prospective oriented research? Miner. Process. Extr. Metall. Rev. 44 (2023) 64–75, https://doi.org/10.1080/08827508.2021.2023520.
[6] K. Prebanić, M. Vukomanović, Realizing the need for digital transformation of stakeholder management: a systematic review in the construction industry, Sustainability 13 (2021) 12690, https://doi.org/10.3390/su132212690.
[7] D. Peral-García, J. Cruz-Benito, F.J. García-Peñalvo, Comparing natural language processing and quantum natural processing approaches in text classification tasks, Expert Syst. Appl. 254 (2024) 124427, https://doi.org/10.1016/j.eswa.2024.124427.
[8] A. Waqar, K. Skrzypkowski, H. Almujibah, K. Zagórski, M.B. Khan, A. Zagórska, O. Benjeddou, Success of implementing cloud computing for smart development in small construction projects, Appl. Sci. 13 (2023), https://doi.org/10.3390/app13095713.

[9] T.V. Shepitko, S.Y. Lutskiy, A.Y. Landsman, I.A. Artyushenko, Features of the construction of the roadbed in the areas of high-temperature permafrost of the Northern Latitudinal passage, in: S. A.V. (Ed.), AIP Conf. Proc., American Institute of Physics Inc., Russian University of Transport, Russian Federation, Moscow, 2023 020025, https://doi.org/10.1063/5.0103111.
[10] R. Panchendrarajan, A. Zubiaga, Synergizing machine learning & symbolic methods: a survey on hybrid approaches to natural language processing, Expert Syst. Appl. 251 (2024) 124097, https://doi.org/10.1016/j.eswa.2024.124097.
[11] H. Wang, J. Li, H. Wu, E. Hovy, Y. Sun, Pre-trained language models and their applications, Engineering (2022), https://doi.org/10.1016/j.eng.2022.04.024.
[12] B. Shuai, A rationale-augmented NLP framework to identify unilateral contractual change risk for construction projects, Comput. Ind. 149 (2023) 103940, https://doi.org/10.1016/j.compind.2023.103940.
[13] P.V. Thayyib, R. Mamilla, M. Khan, H. Fatima, M. Asim, I. Anwar, M. K. Shamsudheen, M.A. Khan, State-of-the-art of artificial intelligence and big data analytics reviews in five different domains: a bibliometric summary, Sustainability 15 (2023) 4026, https://doi.org/10.3390/su15054026.
[14] Y.-L. Du, T.-H. Yi, X.-J. Li, X.-L. Rong, L.-J. Dong, D.-W. Wang, Y. Gao, Z. Leng, Advances in Intellectualization of Transportation Infrastructures, Engineering 24 (2023) 239–252, https://doi.org/10.1016/j.eng.2023.01.011.
[15] M.Á. Tovar Onofre, J.E. Camargo, Automatic class extraction from spanish text of user stories using natural language processing, in: N. F.R., U. F., S.-G. J.P., B.-F. T. F. (Eds.), Commun. Comput. Inf. Sci., Springer Science and Business Media Deutschland GmbH, UnSecure Lab Research Group, Universidad Nacional de Colombia, Bogotá, Colombia, 2023, pp. 33–47, https://doi.org/10.1007/978-3-031-32213-6_3.
[16] A. Waqar, I. Othman, J.C. Pomares, Impact of 3D printing on the overall project success of residential construction projects using structural equation modelling, Int. J. Environ. Res. Public Health (2023), https://doi.org/10.3390/ijerph20053800.
[17] F. Hussain, S. Mehta, M. Soy, J. Zhang, Natural language processing for construction management: a literature review, Constr. Res. Congr. (2024) 607–618, https://doi.org/10.1061/9780784485262.062.
[18] F. Tao, Q. Qi, L. Wang, A.Y.C. Nee, Digital twins and cyber–physical systems toward smart manufacturing and industry 4.0: correlation and comparison, Engineering 5 (2019) 653–661, https://doi.org/10.1016/j.eng.2019.01.014.
[19] M. Yin, L. Tang, C. Webster, S. Xu, X. Li, H. Ying, An ontology-aided, natural language-based approach for multi-constraint BIM model querying, J. Build. Eng. 76 (2023) 107066, https://doi.org/10.1016/j.jobe.2023.107066.
[20] J. Allen, S. Reddivari, Efficient Parallel Wikipedia Internal Link Extraction for NLP-Assisted Requirements Understanding, in: V.L. H., S. S.S., T. Y., C. A., K. H., T. D., Y. J.-J., S. H. (Eds.), 2022 IEEE 46th Annu. Comput. Software, Appl. Conf, IEEE, School of Computing, University of North Florida, Jacksonville, United States, 2022, pp. 434–435, https://doi.org/10.1109/COMPSAC54236.2022.00077.
[21] T. Bikaun, M. Stewart, M. Hodkiewicz, Using context-free grammar to generate synthetic technical short texts, in: A. H., C. D., F. T. (Eds.), Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), Springer Science and Business Media Deutschland GmbH, Department of Computer Science and Software Engineering, University of Western Australia, Perth, 6009, Australia, 2022, pp. 325–338, https://doi.org/10.1007/978-3-031-22695-3_23.
[22] Z.Z. Chen, M. Yan, X. Xia, Z.X. Liu, Z. Xu, Y. Lei, Research progress of code naturalness and its application, Ruan Jian Xue Bao J. Softw. (2022), https://doi.org/10.13328/j.cnki.jos.006355.
[23] S.-W. Choi, E.-B. Lee, Contractor's risk analysis of engineering procurement and construction (EPC) contracts using ontological semantic model and Bi-long short-term memory (LSTM) technology, Sustainability 14 (2022) 6938, https://doi.org/10.3390/su14116938.
[24] L. De Bruyne, P. Atanasova, I. Augenstein, Joint emotion label space modeling for affect lexica, Comput. Speech Lang. 71 (2022) 101257, https://doi.org/10.1016/j.csl.2021.101257.
[25] C. Wu, X. Li, Y. Guo, J. Wang, Z. Ren, M. Wang, Z. Yang, Natural language processing for smart construction: current status and future directions, Autom. Constr. 134 (2022) 104059, https://doi.org/10.1016/j.autcon.2021.104059.
[26] J.A. Fries, L. Weber, N. Seelam, G. Altay, D. Datta, R. Su, S. Garda, S.M.S. Kang, S. Biderman, M. Samwald, S.H. Bach, W. Kusa, S. Cahyawijaya, F. Barth, S. Ott, M. Sänger, B. Wang, A. Callahan, D.L. Periñán, T. Gigant, P. Haller, J. Chim, J. Posada, J. Giorgi, K.R. Sivaraman, M. Pàmies, M. Nezhurina, R. Martin, M. Freidank, N. Dahlberg, S. Mishra, S. Bose, N. Broad, Y. Labrak, S.S. Deshmukh, S. Kiblawi, A. Singh, M.C. Vu, T. Neeraj, J. Golde, A.V. del Moral, B. Beilharz, BIGBIO: a framework for data-centric biomedical natural language processing, in: K. S., M. S., A. A., B. D., C. K., O. A. (Eds.), Adv. Neural Inf. Process. Syst., Neural information processing systems foundation, Stanford University, United States, 2022.
[27] M. Fan, D. Su, M.W. Bhatt, A. Mangal, Study on non-linear planning model of green building energy consumption under multi-objective optimization, Int. J. Syst. Assur. Eng. Manag 13 (2022) 437–443, https://doi.org/10.1007/s13198-021-01459-3.
[28] M. Geva, A. Caciularu, K.R. Wang, Y. Goldberg, Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space, in: G. Y., K. Z., Z. Y. (Eds.), Proc. 2022 Conf. Empir. Methods Nat. Lang. Process. EMNLP 2022, Association for Computational Linguistics (ACL), Allen Institute for AI, 2022, pp. 30–45.
[29] K. Gimasheva, E. Gladkikh, S.I. Chuprina, An ontology-driven methods unified application in text data processing problems, Вестник ПерМского Университета МатеМатика Механика ИнфорМатика (2022) 49–58, https://doi.org/10.17072/1993-0550-2022-3-49-58.
[30] A. Singh, Integrating deep learning techniques in information retrieval: a hybrid approach to relevance optimization, J. Inf. Syst. Eng. Manag 10 (2025) 311–316, https://doi.org/10.52783/jisem.v10i53s.10876.
[31] T.-H. Lin, Y.-H. Huang, A. Putranto, Intelligent question and answer system for building information modeling and artificial intelligence of things based on the bidirectional encoder representations from transformers model, Autom. Constr. 142 (2022) 104483, https://doi.org/10.1016/j.autcon.2022.104483.
[32] A. Kuang, Construction of personalized advertising accuracy model based on artificial intelligence, in: 2022 Int. Conf. Artif. Intell. Auton. Robot Syst, IEEE, 2022, pp. 395–398, https://doi.org/10.1109/AIARS57204.2022.00095.
[33] Y. Lu, J. Zhang, Bibliometric analysis and critical review of the research on big data in the construction industry, Eng. Constr. Archit. Manag 29 (2022) 3574–3592, https://doi.org/10.1108/ECAM-01-2021-0005.
[34] M. Locatelli, L.C. Tagliabue, G.M. Di Giuda, ArchiBERTo: a Hierarchization Quality Objectives NLP Tool in the Italian Architecture, Engineering and Construction Sector, in: L. P., B. P., S. L., T. V., D.C. M., L. N. (Eds.), CEUR Workshop Proc., CEUR-WS, Politecnico di Milano, Department of Architecture, Built Environment and Construction Engineering, Milan, 20133, Italy, 2022: pp. 8–25.
[35] P.M. Mah, I. Skalna, J. Muzam, Natural language processing and artificial intelligence for enterprise management in the era of industry 4.0, Appl. Sci. 12 (2022) 9207, https://doi.org/10.3390/app12189207.
[36] F. Ortin, J. Quiroga, O. Rodriguez-Prieto, M. Garcia, An empirical evaluation of Lex/Yacc and ANTLR parser generation tools, PLoS One 17 (2022) e0264326, https://doi.org/10.1371/journal.pone.0264326.
[37] L. P, B. P, S. L, T. V, D.C. M, L. N, eds., AIxPA 2022 - Proceedings of 1st Workshop on AI for Public Administration, co-located with 21st International Conference of the Italian Association for Artificial Intelligence, AIxIA 2022, in: CEUR Workshop Proc., CEUR-WS, 2022.
[38] N. S, S. J, eds., ICBASE 2022 - Proceedings of the 2022 3rd International Conference on Big Data and Artificial Intelligence and Software Engineering, in: CEUR Workshop Proc., CEUR-WS, 2022.
[39] R. Saini, G. Mussbacher, J.L.C. Guo, J. Kienzle, Automated, interactive, and traceable domain modelling empowered by artificial intelligence, Softw. Syst. Model 21 (2022) 1015–1045, https://doi.org/10.1007/s10270-021-00942-6.
[40] Q. Shen, S. Wu, Y. Deng, H. Deng, J.C.P. Cheng, BIM-based dynamic construction safety rule checking using ontology and natural language processing, Buildings (2022), https://doi.org/10.3390/buildings12050564.
[41] Z. Li, Z. Wang, W. Wang, K. Hung, H. Xie, F.L. Wang, Retrieval-augmented generation for educational application: a systematic survey, Comput. Educ. Artif. Intell. 8 (2025) 100417, https://doi.org/10.1016/j.caeai.2025.100417.
[42] D. Krütli, T. Hanne, Augmenting LLMs to securely retrieve information for construction and facility management, Information 16 (2025), https://doi.org/10.3390/info16020076.
[43] Z. Haider, M.H. Rahman, V. Devabhaktuni, S. Moeykens, P. Chakraborty, A framework for mitigating malicious RLHF feedback in LLM training using consensus based reward, Sci. Rep. 15 (2025) 9177, https://doi.org/10.1038/s41598-025-92889-7.
[44] D. Sivan, S. Kumar, V. Raj, R. Jose, 7 Reinforcement Learning from Human Feedback (RLHF), in: 2024: pp. 135–154. ⟨https://doi.org/10.1515/9783111425078-007⟩.
[45] L. Höltgen, S. Zentgraf, P. Hagedorn, M. König, Utilizing large language models for semantic enrichment of infrastructure condition data: a comparative study of GPT and Llama models, AI Civ. Eng. 4 (2025) 14, https://doi.org/10.1007/s43503-025-00055-9.
[46] S. Singh, M. Kaur, P. Tanwar, S. Sharma, Design and development of conversational chatbot for Covid-19 using NLP: an AI application, in: 2022 6th Int. Conf. Comput. Methodol. Commun, IEEE, 2022, pp. 1654–1658, https://doi.org/10.1109/ICCMC53470.2022.9753893.
[47] K.D. Strang, Z. Sun, ERP Staff versus AI recruitment with employment real-time big data, Discov. Artif. Intell. 2 (2022) 21, https://doi.org/10.1007/s44163-022-00037-1.
[48] H. Wang, J. Li, H. Wu, E. Hovy, Y. Sun, Pre-trained language models and their applications, Engineering 25 (2023) 51–65, https://doi.org/10.1016/j.eng.2022.04.024.
[49] H. Wang, X. Meng, X. Zhu, Improving knowledge capture and retrieval in the BIM environment: combining case-based reasoning and natural language processing, Autom. Constr. 139 (2022) 104317, https://doi.org/10.1016/j.autcon.2022.104317.
[50] N. Wang, R.R.A. Issa, C.J. Anumba, NLP-Based query-answering system for information extraction from building information models, J. Comput. Civ. Eng. 36 (2022), https://doi.org/10.1061/(ASCE)CP.1943-5487.0001019.
[51] R. Wang, Y. Zhang, S. Mao, Intelligent text classification and knowledge mining of hidden safety hazards in hydropower engineering construction, Shuili Fadian Xuebao J. Hydroelectr. Eng. 41 (2022) 96–106, https://doi.org/10.11660/slfdxb.20221110.
[52] F. Li, H. Li, B. Niu, J. Chen, Privacy computing: concept, computing framework, and future development trends, Engineering 5 (2019) 1179–1192, https://doi.org/10.1016/j.eng.2019.09.002.
[53] M. Zhou, N. Duan, S. Liu, H.-Y. Shum, Progress in neural NLP: modeling, learning, and reasoning, Engineering 6 (2020) 275–290, https://doi.org/10.1016/j.eng.2019.12.014.
[54] Q. Zhao, R. Wang, P. Xu, W. Yang, Construction and application of nlp-based knowledge graph in CNC equipment fault field, in: 2022 13th Int. Conf. Mech.

Aerosp. Eng., IEEE, Avic Manufacturing Technology Institute, Information Center, Beijing, China, 2022, pp. 514–519, https://doi.org/10.1109/ICMAE56000.2022.9852902.
[55] S. Zentgraf, P. Hagedorn, M. König, Multi-requirements ontology engineering for automated processing of document-based building codes to linked building data properties, IOP Conf. Ser. Earth Environ. Sci. 1101 (2022) 092007, https://doi.org/10.1088/1755-1315/1101/9/092007.
[56] Z. Zheng, Y.-C. Zhou, X.-Z. Lu, J.-R. Lin, Knowledge-informed semantic alignment and rule interpretation for automated compliance checking, Autom. Constr. 142 (2022) 104524, https://doi.org/10.1016/j.autcon.2022.104524.
[57] T. Akanbi, J. Zhang, Design information extraction from construction specifications to support cost estimation, Autom. Constr. 131 (2021) 103835, https://doi.org/10.1016/j.autcon.2021.103835.
[58] Y. Zhou, J. She, Y. Huang, L. Li, L. Zhang, J. Zhang, A design for safety (DFS) semantic framework development based on natural language processing (NLP) for automated compliance checking using BIM: the case of China, Buildings 12 (2022) 780, https://doi.org/10.3390/buildings12060780.
[59] J. Bian, Z. Mao, Y. Liu, Y. Ma, W. Liu, B. Jiang, Construction and reasoning method of fault knowledge graph with application of engineering machinery, in: 2021 China Autom. Congr, IEEE, College of Automation Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China, 2021, pp. 2577–2581, https://doi.org/10.1109/CAC53003.2021.9727906.
[60] A. Bhatia, A. Pinto, Automated construction of knowledge-bases for safety critical applications: Challenges and opportunities, in: M. A., H. K., H. K., F. H.-G., G. A., L. D., S. R., van H. F. (Eds.), CEUR Workshop Proc., CEUR-WS, Raytheon Technologies Research Center, 2855 Telegraph Ave, Ste 410, Berkeley, CA, United States, 2021.
[61] M.F. Wong, C.W. Tan, Aligning crowd-sourced human feedback for reinforcement learning on code generation by large language models, IEEE Trans. Big Data PP (2024) 1–12, https://doi.org/10.1109/TBDATA.2024.3524104.
[62] M. Raiaan, S. Mukta, K. Fatema, N. Fahad, S. Sakib, M.M.J. Mim, J. Ahmad, M. E. Ali, S. Azam, A review on large language models: architectures, applications, taxonomies, open issues and challenges, IEEE Access PP 1 (2024), https://doi.org/10.1109/ACCESS.2024.3365742.
[63] Y. Ding, J. Ma, X. Luo, Applications of natural language processing in construction, Autom. Constr. 136 (2022) 104169, https://doi.org/10.1016/j.autcon.2022.104169.
[64] F. ul Hassan, T. Le, X. Lv, Addressing legal and contractual matters in construction using natural language processing: a critical review, J. Constr. Eng. Manag 147 (2021), https://doi.org/10.1061/(ASCE)CO.1943-7862.0002122.
[65] S. Baek, W. Jung, S.H. Han, A critical review of text-based research in construction: Data source, analysis method, and implications, Autom. Constr. 132 (2021) 103915, https://doi.org/10.1016/j.autcon.2021.103915.
[66] E.V. Godinez, Z. Szlávik, E. Contempré, R.-J. Sips, What do you mean, doctor? A knowledge-based approach for word sense disambiguation of medical terminology, in: P. C., F. A., G. H. (Eds.), Heal. 2021 - 14th Int. Conf. Heal. Informatics; Part 14th Int. Jt. Conf. Biomed. Eng. Syst. Technol. BIOSTEC 2021, SciTePress, myTomorrows, Anthony Fokkerweg 61, Amsterdam, 1059CP, Netherlands, 2021: pp. 273–280.
[67] D. Feng, H. Chen, A small samples training framework for deep Learning-based automatic information extraction: case study of construction accident news reports analysis, Adv. Eng. Inform. 47 (2021) 101256, https://doi.org/10.1016/j.aei.2021.101256.
[68] H. Jang, B. Yoon, TechWordNet: Development of semantic relation for technology information analysis using F-term and natural language processing, Inf. Process. Manag. 58 (2021) 102752, https://doi.org/10.1016/j.ipm.2021.102752.
[69] R. Hantach, G. Lechuga, P. Calvez, Key Information Recognition from Piping and Instrumentation Diagrams: Where We Are? in: B.S. E.H., P. U. (Eds.), Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics) Springer Science and Business Media Deutschland GmbH, CSAI Lab ENGIE, Paris, France, 2021, pp. 504–508, https://doi.org/10.1007/978-3-030-86159-9_36.
[70] V. dos Santos, E.F. de Souza, K.R. Felizardo, W.M. Watanabe, A.C. Junior, S. M. Aluisio, N.L. Vijaykumar, Using natural language processing to build graphical abstracts to be used in studies selection activity in secondary studies, in: B. M.T., S. G., S. A. (Eds.), 2021 47th Euromicro Conf. Softw. Eng. Adv. Appl, IEEE, University of São Paulo (ICMC/USP), São Carlos, Brazil, 2021, pp. 1–8, https://doi.org/10.1109/SEAA53835.2021.00009.
[71] S. Shin, C. Lee, R.R.A. Issa, Advanced BIM Platform Based on the Spoken Dialogue for End-User. Lect. Notes Civ. Eng., 2021, pp. 123–132, https://doi.org/10.1007/978-3-030-51295-8_11.
[72] M.-J. Park, E.-B. Lee, S.-Y. Lee, J.-H. Kim, A digitalized design risk analysis tool with machine-learning algorithm for EPC contractor's technical specifications assessment on bidding, Energies 14 (2021) 5901, https://doi.org/10.3390/en14185901.
[73] Y. Pan, L. Zhang, Roles of artificial intelligence in construction engineering and management: a critical review and future trends, Autom. Constr. 122 (2021) 103517, https://doi.org/10.1016/j.autcon.2020.103517.
[74] M. Locatelli, E. Seghezzi, L. Pellegrini, L. Tagliabue, G. Di Giuda, Exploring natural language processing in construction and integration with building information modeling: a scientometric analysis, Buildings 11 (2021) 583, https://doi.org/10.3390/buildings11120583.
[75] M. Locatelli, E. Seghezzi, G.M. Di Giuda, Exploring bim and nlp applications: a scientometric approach, Proc. Int. Struct. Eng. Constr. 8 (2021), https://doi.org/10.14455/ISEC.2021.8(1).CON-12.
[76] S. Wu, X. Li, S. Zhu, W. Shen, J. Wu, Y. Sun, Application of Artificial Intelligence in Teaching Reform of New Economics, in: 2021 3rd Int. Conf. Artif. Intell. Adv. Manuf., ACM, New York, NY, USA, Hebei University of Economics and Business, Shijiazhuang, 050061, China, 2021: pp. 572–577. ⟨https://doi.org/10.1145/3495018.3495121⟩.
[77] L.W.Y. Yang, W.Y. Ng, L.L. Foo, Y. Liu, M. Yan, X. Lei, X. Zhang, D.S.W. Ting, Deep learning-based natural language processing in ophthalmology: applications, challenges and future directions, Curr. Opin. Ophthalmol. 32 (2021) 397–405, https://doi.org/10.1097/ICU.0000000000000789.
[78] M.H. Kazemi, A. Alvanchi, Application of NLP-based models in automated detection of risky contract statements written in complex script system, Expert Syst. Appl. 259 (2025) 125296, https://doi.org/10.1016/j.eswa.2024.125296.
[79] I. Dikmen, G. Eken, H. Erol, M.T. Birgonul, Automated construction contract analysis for risk and responsibility assessment using natural language processing and machine learning, Comput. Ind. 166 (2025) 104251, https://doi.org/10.1016/j.compind.2025.104251.
[80] H. Zhang, Neural network-based tree translation for knowledge base construction, IEEE Access 9 (2021) 38706–38717, https://doi.org/10.1109/ACCESS.2021.3063234.
[81] D.N.A, Akash Roshan, Meenu Garg, Intelligent voice assisstant for desktop using NLP and AI, Int. J. Mod. Trends Sci. Technol. 6 (2020) 328–331, https://doi.org/10.46501/IJMTST061261.
[82] M. Budnikov, A. Bykova, I.P. Yamshchikov, Generalization potential of large language models, Neural Comput. Appl. 37 (2025) 1973–1997, https://doi.org/10.1007/s00521-024-10827-6.
[83] K. González Barman, S. Lohse, H.W. de Regt, Reinforcement learning from human feedback in LLMs: whose culture, whose values, whose perspectives? Philos. Technol. 38 (2025) 35, https://doi.org/10.1007/s13347-025-00861-0.
[84] C.-L. Chou, C.-H. Chang, Y.-H. Lin, K.-C. Chien, On the construction of web ner model training tool based on distant supervision, ACM Trans. Asian Low. Resour. Lang. Inf. Process 19 (2020) 1–28, https://doi.org/10.1145/3422817.
[85] N. Dehbozorgi, M. Lou Maher, M. Dorodchi, Sentiment Analysis on Conversations in Collaborative Active Learning as an Early Predictor of Performance, in: 2020 IEEE Front. Educ. Conf, IEEE, University of North Carolina at Charlotte, Dept. of Computer Science, Charlotte, United States, 2020, pp. 1–9, https://doi.org/10.1109/FIE44824.2020.9274119.
[86] G.M. Di Giuda, M. Locatelli, E. Seghezzi, NLP And BIM in AECO sector: a state of the art, Proc. Int. Struct. Eng. Constr. 7 (2020), https://doi.org/10.14455/ISEC.2020.7(2).CON-22.
[87] G.M. Di Giuda, M. Locatelli, E. Seghezzi, Natural language processing and bim in aeco sector: a state of the art, Proc. Int. Struct. Eng. Constr. 7 (2020), https://doi.org/10.14455/ISEC.2020.7(2).CON-22.
[88] İ. Baydili, B. Tasci, G. Tasci, Deep learning-based detection of depression and suicidal tendencies in social media data with feature selection, Behav. Sci. 15 (2025), https://doi.org/10.3390/bs15030352.
[89] M. A, W. W, eds., 8th International Conference on Design, User Experience, and Usability, DUXU 2019, held as part of the 21st International Conference on Human-Computer Interaction, HCI International 2019, Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics) 11586 LNCS (2019).
[90] X.-G. Zhou, R.-B. Gong, F.-G. Shi, Z.-F. Wang, PetroKG: construction and application of knowledge graph in upstream area of petroChina, J. Comput. Sci. Technol. 35 (2020) 368–378, https://doi.org/10.1007/s11390-020-9966-7.
[91] C. Abderrouaf, M. Oussalah, On Online Hate Speech Detection. Effects of Negated Data Construction, in: B. C., H. J., K. L., H. X.T., A. R., T. Y., B. R., Z. C., L. K., Y. Y. F. (Eds.), 2019 IEEE Int. Conf. Big Data (Big Data), IEEE, University of Oulu, Faculty of Information Technology, CMVS, Oulu, 90014, Finland, 2019: pp. 5595–5602. ⟨https://doi.org/10.1109/BigData47090.2019.9006336⟩.
[92] C. Arora, M. Sabetzadeh, S. Nejati, L. Briand, An active learning approach for improving the accuracy of automated domain model extraction, ACM Trans. Softw. Eng. Method. 28 (2019) 1–34, https://doi.org/10.1145/3293454.
[93] P. Bob, Impact of artificial intelligence on management of large complex projects, Electron. J. Bus. Ethics Organ. Stud. (2019).
[94] C. Anuradha, N. Priya, S. Sangeetha, G. Kavitha, AI NLP chatbot for HR and employee support, Int. J. Innov. Technol. Explor. Eng. 8 (2019) 580–582, https://doi.org/10.35940/ijitee.I3115.0789S319.
[95] H. Chen, X. Luo, An automatic literature knowledge graph and reasoning network modeling framework based on ontology and natural language processing, Adv. Eng. Inform. 42 (2019) 100959, https://doi.org/10.1016/j.aei.2019.100959.
[96] C. Chiarcos, C. Fäth, Graph-based annotation engineering: Towards a gold corpus for role and reference grammar, in: E. M., de M. G., F. C., M. J.P., B. P., C. C., K. B., D. M. (Eds.), OpenAccess Ser. Informatics, Schloss Dagstuhl- Leibniz-Zentrum fur Informatik GmbH, Dagstuhl Publishing, Applied Computational Linguistics Lab, Goethe University, Frankfurt, Germany, 2019, https://doi.org/10.4230/OASIcs.LDK.2019.9.
[97] D.N. Chuong, P. Seresangtakul, Semi-Automatic Word-Aligned Tool for Thai-Vietnamese Parallel Corpus Construction, in: 2019 16th Int. Jt. Conf. Comput. Sci. Softw. Eng., IEEE, Natural Language and Speech Processing Laboratory, Khon Kaen University, Khon Kaen, Thailand, 2019, pp. 121–125, https://doi.org/10.1109/JCSSE.2019.8864191.
[98] H. Dawood, J. Siddle, N. Dawood, Integrating IFC and NLP for automating change request validations, J. Inf. Technol. Constr. 24 (2019) 540–552, https://doi.org/10.36680/j.itcon.2019.030.
[99] B. Galitsky, Dev. Enterp. Chatbots (2019), https://doi.org/10.1007/978-3-030-04299-8.

[100] T. Hachaj, M.R. Ogiela, What Can Be Learned from Bigrams Analysis of Messages in Social Network?, in: L. Q., L. W., W. L. (Eds.), 2018 11th Int. Congr. Image Signal Process. Biomed. Eng. Informatics, IEEE, Institute of Computer Science, Pedagogical University of Cracow, Cracow, Poland, 2018: pp. 1–4. ⟨https://doi.org/10.1109/CISP-BMEI.2018.8633108⟩.
[101] E. Hjelseth, Foundations for BIM-based model checking systems: transforming regulations into computable rules in BIM-based model checking systems, 1 b. (Flere Pag.) (2019).
[102] N. Jung, G. Lee, Automated classification of building information modeling (BIM) case studies by BIM use based on natural language processing (NLP) and unsupervised learning, Adv. Eng. Inform. 41 (2019) 100917, https://doi.org/10.1016/j.aei.2019.04.007.
[103] J. Krishnan, P. Coronado, T. Reed, Seva: A systems engineer's virtual assistant, in: C. P., G. A., van H. F., M. A., H. K., L. D. (Eds.), CEUR Workshop Proc., CEUR-WS, Department of Computer Science, George Mason University, United States, 2019.
[104] Y. Kim, J. Lee, E.-B. Lee, J.-H. Lee, Application of natural language processing (NLP) and text-mining of big-data to engineering-procurement-construction (EPC) bid and contract documents. 2020 6th Conf. Data Sci. Mach. Learn. Appl, IEEE, 2020, pp. 123–128, https://doi.org/10.1109/CDMA47397.2020.00027.
[105] J. Li, K. Afsari, N. Li, J. Peng, Z. Wu, H. Cui, A review for presenting building information modeling education and research in China, J. Clean. Prod. 259 (2020) 120885, https://doi.org/10.1016/j.jclepro.2020.120885.
[106] T. Louge, M.H. Karray, B. Archimede, Investigating a Method for Automatic Construction and Population of Ontologies for Services: Performances and Limitations, in: 2018 IEEE/ACS 15th Int. Conf. Comput. Syst. Appl, IEEE, Université de Toulouse, Institut de Recherche en Astrophysique et Planetologie, Tarbes, France, 2018, pp. 1–6, https://doi.org/10.1109/AICCSA.2018.8612844.
[107] A.B. Saka, L.O. Oyedele, L.A. Akanbi, S.A. Ganiyu, D.W.M. Chan, S.A. Bello, Conversational artificial intelligence in the AEC industry: a review of present status, challenges and opportunities, Adv. Eng. Inform. 55 (2023) 101869, https://doi.org/10.1016/j.aei.2022.101869.
[108] R. Nidhi, D.T. Singh, D.D. Lobiyal, MT Embedded E-Learning in India - Challenges for NLP/AI, Int. J. Eng. Adv. Technol. 9 (2019) 2577–2582, https://doi.org/10.35940/ijeat.A9348.109119.
[109] B. Prieto, Impact of artificial intelligence on management of large complex projects, Electron. J. Bus. Ethics Organ. Stud. (2019).
[110] S. Ganesh, A.K. Talukder, Formal Methods, artificial intelligence, big-data analytics, and knowledge engineering in medical care to reduce disease burden and health disparities. Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), 2018, https://doi.org/10.1007/978-3-030-04780-1_21.
[111] S. Ganesh, A.K. Talukder, Formal Methods, Artificial Intelligence, Big-Data Analytics, and Knowledge Engineering in Medical Care to Reduce Disease Burden and Health Disparities, in: BMC Bioinformatics, Computer Science and Engineering Department, Oakland University, 2200 N. Squirrel Rd, Rochester, 48309, MI, United States, 2018: pp. 307–321. ⟨https://doi.org/10.1007/978-3-030-04780-1_21⟩.
[112] M. Arias, A. Buccella, A. Cechich, A framework for managing requirements of software product lines, Electron. Notes Theor. Comput. Sci. 339 (2018) 5–20, https://doi.org/10.1016/j.entcs.2018.06.002.
[113] E. Nazaruka, J. Osis, Determination of Natural Language Processing Tasks and Tools for Topological Functioning Modelling, in: D. E., S. G., M. L., M. L. (Eds.), Proc. 13th Int. Conf. Eval. Nov. Approaches to Softw. Eng., SCITEPRESS - Science and Technology Publications, Department of Applied Computer Science, Riga Technical University, Sētas iela 1, Riga, LV-1048, Latvia, 2018: pp. 501–512. ⟨https://doi.org/10.5220/0006817205010512⟩.
[114] S. Dutta, B. Gunay, S. Bucking, Benchmarking operational performance of buildings by text mining tenant surveys, Sci. Technol. Built Environ. 27 (2020) 1–20, https://doi.org/10.1080/23744731.2020.1851545.
[115] L. Bharadwaj, Sentiment analysis in online product reviews: mining customer opinions for sentiment classification, Int. J. Multidiscip. Res. 5 (2023), https://doi.org/10.36948/ijfmr.2023.v05i05.6090.
[116] A. Chokor, H. Naganathan, W.K. Chong, M.El Asmar, Analyzing arizona OSHA injury reports using unsupervised machine learning, Procedia Eng. 145 (2016) 1588–1593, https://doi.org/10.1016/j.proeng.2016.04.200.
[117] T. Kostareva, S. Chuprina, A. Nam, Using ontology-driven methods to develop frameworks for tackling NLP problems, in: I. D.I., K. M.Y., N. S., P. A., V. K.V., L. N., P. A., S. A.V., L. V.G. (Eds.), CEUR Workshop Proc., CEUR-WS, Perm State University, 15 Bukireva st., Perm, 614068, Russian Federation, 2016: pp. 102–113.
[118] A.J.-P. Tixier, M.R. Hallowell, B. Rajagopalan, D. Bowman, Automated content analysis for construction safety: a natural language processing system to extract precursors and outcomes from unstructured injury reports, Autom. Constr. 62 (2016) 45–56, https://doi.org/10.1016/j.autcon.2015.11.001.
[119] A.J.-P. Tixier, M.R. Hallowell, B. Rajagopalan, D. Bowman, Automated content analysis for construction safety: a natural language processing system to extract precursors and outcomes from unstructured injury reports, Autom. Constr. 62 (2016) 45–56, https://doi.org/10.1016/j.autcon.2015.11.001.
[120] Y. Shu, Y.H. Lun, Y.X. Run, W. Ye, An automated method for constructing ontology, in: B. M.S.P., W. L. (Eds.), 2016 7th IEEE Int. Conf. Softw. Eng. Serv. Sci., IEEE, Department of Computer Science and Engineering, Xin Jiang Agriculture University, XJAU, NongDa east road, Urumqi, 830052, China, 2016: pp. 538–541. ⟨https://doi.org/10.1109/ICSESS.2016.7883127⟩.
[121] N. Arman, S. Jabbarin, Generating use case models from arabic user requirements in a semiautomated approach using a natural language processing tool, J. Intell. Syst. 24 (2015) 277–286, https://doi.org/10.1515/jisys-2014-0092.
[122] M. Landhausser, S.J. Korner, W.F. Tichy, J. Keim, J. Krisch, DeNom: a tool to find problematic nominalizations using NLP, in: H. R., B. N., C.-H. J., G. J. (Eds.), 2015 IEEE Second Int. Work. Artif. Intell. Requir. Eng., IEEE, Karlsruhe Institute of Technology, Germany, 2015: pp. 1–8. ⟨https://doi.org/10.1109/AIRE.2015.7337623⟩.
[123] L. Lin, H. Chen, Y. Bi, The Designing and Construction of Domain-oriented Vietnamese-English-Chinese FrameNet, in: S. M., L. Z., L. Y., Z. M. (Eds.), Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), Springer Verlag, Luoyang University of Foreign Languages, Luoyang, Henan, 471003, China, 2015: pp. 53–65. ⟨https://doi.org/10.1007/978-3-319-25816-4_5⟩.
[124] J. Wao, A review of the value engineering methodology: Limitations and solutions for sustainable construction, in: SAVE Value Summit 2015, SAVE International, City of Gainesville Public Works Department, Gainesville, FL, United States, 2015: pp. 1–7.
[125] M. Yalcinkaya, V. Singh, Patterns and trends in building information modeling (BIM) research: a latent semantic analysis, Autom. Constr. 59 (2015) 68–80, https://doi.org/10.1016/j.autcon.2015.07.012.
[126] L. Abouenour, M. Nasri, K. Bouzoubaa, A. Kabbaj, P. Rosso, Construction of an ontology for intelligent Arabic QA systems leveraging the conceptual graphs representation, J. Intell. Fuzzy Syst. 27 (2014) 2869–2881, https://doi.org/10.3233/IFS-141248.
[127] C. Arora, M. Sabetzadeh, L. Briand, F. Zimmer, Improving requirements glossary construction via clustering, in: Proc. 8th ACM/IEEE Int. Symp. Empir. Softw. Eng. Meas., ACM, New York, NY, USA, Reliability and Trust, University of Luxembourg, Luxembourg, 2014: pp. 1–10. ⟨https://doi.org/10.1145/2652524.2652530⟩.
[128] S. Jabbarin, N. Arman, Constructing use case models from Arabic user requirements in a semi-automated approach, in: 2014 World Congr. Comput. Appl. Inf. Syst, IEEE, Department of Computer Science and Engineering, Palestine Polytechnic University Hebron, Palestine, 2014, pp. 1–4, https://doi.org/10.1109/WCCAIS.2014.6916558.
[129] D. HUANG, S. WANG, F. REN, Creating Chinese-english comparable corpora, IEICE Trans. Inf. Syst. E96.D (2013) 1853–1861, https://doi.org/10.1587/transinf.E96.D.1853.
[130] J. Kwon, B. Kim, S. Lee, H. Kim, Automated procedure for extracting safety regulatory information using natural language processing techniques and ontology, in: K. C.J., B. C. (Eds.), Proceedings, Annu. Conf. - Can. Soc. Civ. Eng., Canadian Society for Civil Engineering, Department of Civil and Environmental Engineering, University of Michigan, United States, 2013: pp. 1213–1220.
[131] D. Schwab, L.T. Lim, Blexisma2: A distributed agent framework for constructing a semantic lexical database based on conceptual vectors, in: 2008 First Int. Conf. Distrib. Framew. Appl., IEEE, Laboratoire d'Informatique de Grenoble, Universite Pierre Mendes France, Grenoble 2, France, 2008: pp. 102–110. ⟨https://doi.org/10.1109/ICDFMA.2008.4784421⟩.
[132] Y. Wilks, Wordnet as a resource for NLP: Yorick Wilks' reflections on the ontological debate—and how to cope with ambiguity and vagueness of natural language, Int. J. Speech Technol. 11 (2008) 109–119, https://doi.org/10.1007/s10772-009-9045-5.
[133] M.G. Ilieva, Graphical notation for natural language and knowledge representation, in: 19th Int. Conf. Softw. Eng. Knowl. Eng. SEKE 2007, Dept. of Computer Science and Software Engineering, Concordia University, Montreal, Canada, 2007: pp. 361–367.
[134] F. Roulland, A. Kaplan, S. Castellani, C. Roux, A. Grasso, K. Pettersson, J. O'Neill, Query reformulation and refinement using nlp-based sentence clustering. Adv. Inf. Retr, Springer Berlin Heidelberg, Berlin, Heidelberg, Grenoble, France, 2007, pp. 210–221, https://doi.org/10.1007/978-3-540-71496-5_21.
[135] Y. Wilks, Ontotherapy, or how to stop worrying about what there is, in: Proc. 4th Int. Work. Nat. Lang. Process. Cogn. Sci. - NLPCS 2007; Conjunction with ICEIS 2007, University of Sheffield, Department of Computer Science, Sheffield, S1 4DP, Regent Court, 211 Portobello Street, United Kingdom, 2007: pp. 3–22.
[136] Y. Liu, Z.-F. Sui, Y.-W. Hu, T.-L. Ji, Automatic construction on of domain ontology, Beijing Youdian Daxue, Xuebao/J. Beijing Univ. Posts Telecom 29 (2006) 65–69.
[137] D. Shepherd, L. Pollock, K. Vijay-Shanker, Towards supporting on-demand virtual remodularization using program graphs, in: Proc. 5th Int. Conf. Asp. Softw. Dev., ACM, New York, NY, USA, University of Delaware, Newark, DE 19816, United States, 2006: pp. 3–14. ⟨https://doi.org/10.1145/1119655.1119660⟩.
[138] W. Zhou, Z. Liu, Y. Zhao, L. Xu, G. Chen, Q. Wu, M. Huang, Y. Qiang, A Semi-automatic Ontology Learning Based on WordNet and Event-based Natural Language Processing, in: 2006 Int. Conf. Inf. Autom., IEEE, School of Computer Engineering and Science, Shanghai University, Shanghai, China, 2006: pp. 240–244. ⟨https://doi.org/10.1109/ICINFA.2006.374119⟩.
[139] M E., C. Laboratoire CEDRIC 292 rue Saint Martin, Paris cedex 3, 75141, M. F., S. of C. University of Salford Science and Engineering, Newton Building, Salford, M5 4WT, eds., 9th International Conference on Application of Natural Language to Information Systems, NLDB 2004, Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics) 3136 (2004) 1–435.
[140] B. D.G., D. of C.S. University College Cork The Kane Building, College Road, Cork, A. K.D., L.R. and D.C. University of Pittsburgh 3939 O'Hara Street, Pittsburgh, 15260, PA, eds., 5th International Conference on Case-Based Reasoning, ICCBR 2003, Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics) 2689 (2003) 1–733.
[141] Jun Zhao, Zhifang Sui, Bridge knowledge and languages: the application of computational linguistics, in: Z. C. (Ed.), Int. Conf. Nat. Lang. Process. Knowl. Eng. 2003. Proceedings. 2003, IEEE, National Key Lab of Pattern Recognition,

Institute of Automation, Chinese Academy of Sciences, China, 2003, pp. 828–837, https://doi.org/10.1109/NLPKE.2003.1276020.
[142] A. Ranta, K. Angelov, N. Gruzitis, P. Kolachina, Abstract syntax as interlingua: scaling up the grammatical framework from controlled languages to robust pipelines, Comput. Linguist 46 (2020) 425–486, https://doi.org/10.1162/coli_a_00378.
[143] S. Das, Z. Ali, S.-N. Bandi, A. Bhagat, N. Chandrakumar, P. Kucheria, M. Pariente, A. Singh, B. Tipping, Agile systems engineering in building complex AI systems, in: L. W.F., L. J., S. D.A., M. R. (Eds.), Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), Springer Science and Business Media Deutschland GmbH, AdventHealth Consumer Analytics Team, Orlando, FL, United States, 2021, pp. 192–208, https://doi.org/10.1007/978-3-030-89385-9_12.
[144] Y. Fu, T. Bauer, J. Mostafa, M. Palakal, S. Mukhopadhyay, Concept extraction and association from cancer literature. Proc. 4th Int. Work. Web Inf. Data Manag., ACM, New York, NY, USA, Indiana University, Bloomington, United States, 2002, pp. 100–103, https://doi.org/10.1145/584931.584953.
[145] W. Nam, B. Jang, A survey on multimodal bidirectional machine learning translation of image and natural language processing, Expert Syst. Appl. 235 (2024) 121168, https://doi.org/10.1016/j.eswa.2023.121168.
[146] V. Giordano, G. Puccetti, F. Chiarello, T. Pavanello, G. Fantoni, Unveiling the inventive process from patents by extracting problems, solutions and advantages with natural language processing, Expert Syst. Appl. 229 (2023) 120499, https://doi.org/10.1016/j.eswa.2023.120499.
[147] F. Almusharraf, J. Rose, P. Selby, Engaging unmotivated smokers to move toward quitting: design of motivational interviewing–based chatbot through iterative interactions, J. Med. Internet Res. 22 (2020) e20251, https://doi.org/10.2196/20251.
[148] Y. An, H. Li, T. Su, Y. Wang, Determining uncertainties in AI applications in AEC sector and their corresponding mitigation strategies, Autom. Constr. 131 (2021) 103883, https://doi.org/10.1016/j.autcon.2021.103883.
[149] S. Raza, M. Garg, D.J. Reji, S.R. Bashir, C. Ding, Nbias: a natural language processing framework for BIAS identification in text, Expert Syst. Appl. 237 (2024) 121542, https://doi.org/10.1016/j.eswa.2023.121542.
[150] M. Li, D. Tian, Y. Shen, J. Shi, S. Han, An intelligent extraction and analysis approach of professional technical words for hydraulic engineering by improved Word2vec technology with Attention mechanism, Shuili Xuebao J. Hydraul. Eng. 51 (2020) 816–826, https://doi.org/10.13243/j.cnki.slxb.20190920.
[151] L. Burgueño, R. Clarisó, S. Gérard, S. Li, J. Cabot, An NLP-based architecture for the autocompletion of partial domain models, in: L.R. M., S. S., T. E. (Eds.), Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), Springer Science and Business Media Deutschland GmbH, Open University of Catalonia, Barcelona, Spain, 2021, pp. 91–106, https://doi.org/10.1007/978-3-030-79382-1_6.
[152] X. Huo, Y. Yin, L. Jiao, Y. Zhang, A data-driven and knowledge graph-based analysis of the risk hazard coupling mechanism in subway construction accidents, Reliab. Eng. Syst. Saf. 250 (2024) 110254, https://doi.org/10.1016/j.ress.2024.110254.
[153] F. Kong, S. Ahn, Use of knowledge graphs for construction safety management: a systematic literature review, Information 15 (2024), https://doi.org/10.3390/info15070390.
[154] J. Lin, Z. Hu, J. Zhang, BIM oriented intelligent data mining and representation, Proc. 30th CIB W78 Int. Conf. (2013).
[155] A. Zahedi, J. Abualdenien, F. Petzold, A. Borrmann, BIM-based design decisions documentation using design episodes, explanation tags, and constraints, J. Inf. Technol. Constr. 27 (2022) 756–780, https://doi.org/10.36680/j.itcon.2022.037.
[156] Á. Aldunate, S. Maldonado, C. Vairetti, G. Armelini, Understanding customer satisfaction via deep learning and natural language processing, Expert Syst. Appl. 209 (2022) 118309, https://doi.org/10.1016/j.eswa.2022.118309.
[157] M. KATAOKA, Automatic conversion from building codes into conformance checking computer programs, AIJ J. Technol. Des. 16 (2010) 1193–1196, https://doi.org/10.3130/aijt.16.1193.
[158] G. Sierra, L. Hernández, Automatic construction of the knowledge base of an onomasiological dictionary, in: T. L., A. S., S. S., H. Y., R. A., T. C. (Eds.), CEUR Workshop Proc., CEUR-WS, Language Engineering Group, Engineering Institute Universidad Nacional Autónoma de México, Ciudad Universitaria, México, Mexico, 2013.
[159] J. Zhang, N. El-Gohary, Extraction of Construction Regulatory Requirements from Textual Documents Using Natural Language Processing Techniques, in: Comput. Civ. Eng., American Society of Civil Engineers, Reston, VA, University of Illinois at Urbana-Champaign, Urbana, IL 61801-2352, United States, 2012: pp. 453–460. ⟨https://doi.org/10.1061/9780784412343.0057⟩.
[160] F. Alvarez, A. Vaquero, F. Sáenz, Conceptual modeling of ontology-based linguistic resources with a focus on semantic relations. Proc. 6th Int. Conf. Lang. Resour. Eval. Lr. 2008, European Language Resources Association (ELRA), Universidad Autónoma de Sinaloa, Ángel Flores y Vicente Riva Palacios s/n, Culiacán, Sinaloa, 80000, Mexico, 2008, pp. 879–886.
[161] A. Wen, S. Fu, S. Moon, M. El Wazir, A. Rosenbaum, V.C. Kaggal, S. Liu, S. Sohn, H. Liu, J. Fan, Desiderata for delivering NLP to accelerate healthcare AI advancement and a Mayo Clinic NLP-as-a-service implementation, Npj Digit. Med. 2 (2019) 130, https://doi.org/10.1038/s41746-019-0208-8.
[162] H. Shakil, A. Farooq, J. Kalita, Abstractive text summarization: State of the art, challenges, and improvements, Neurocomputing 603 (2024) 128255, https://doi.org/10.1016/j.neucom.2024.128255.
[163] K.-L. Chen, M.-H. Tsai, Conversation-based information delivery method for facility management, Sensors 21 (2021) 4771, https://doi.org/10.3390/s21144771.
[164] P.F. Christiano, J. Leike, T.B. Brown, M. Martic, S. Legg, D. Amodei, Deep reinforcement learning from human preferences, Adv. Neural Inf. Process. Syst. (2017) 4300–4308.
[165] M.T.R. Laskar, S. Alqahtani, M.S. Bari, M. Rahman, M.A.M. Khan, H. Khan, I. Jahan, M.A.H. Bhuiyan, C.W. Tan, M.R. Parvez, E. Hoque, S. Joty, J.X. Huang, A systematic survey and critical review on evaluating large language models: challenges, limitations, and recommendations. EMNLP 2024 - 2024 Conf. Empir. Methods Nat. Lang. Process. Proc. Conf, 2024, pp. 13785–13816, https://doi.org/10.18653/v1/2024.emnlp-main.764.
[166] K.I. Roumeliotis, N.D. Tselikas, D.K. Nasiopoulos, LLMs in e-commerce: a comparative analysis of GPT and LLaMA models in product review evaluation, Nat. Lang. Process. J. 6 (2024) 100056, https://doi.org/10.1016/j.nlp.2024.100056.
[167] T. Nie, J. Sun, W. Ma, Exploring the roles of large language models in reshaping transportation systems: a survey, framework, and roadmap, Artif. Intell. Transp. 1 (2025) 100003, https://doi.org/10.1016/j.ait.2025.100003.
[168] P.E.D. Love, W. Fang, J. Matthews, S. Porter, H. Luo, L. Ding, Explainable artificial intelligence (XAI): precepts, models, and opportunities for research in construction, Adv. Eng. Inform. 57 (2023) 102024, https://doi.org/10.1016/j.aei.2023.102024.
[169] J. Zhang, M. Xiao, L. Gao, S. Chu, Probability and interval hybrid reliability analysis based on adaptive local approximation of projection outlines using support vector machine, Comput. Civ. Infrastruct. Eng. 34 (2019) 991–1009, https://doi.org/10.1111/mice.12480.
[170] G.G. Tiruneh, A.R. Fayek, V. Sumati, Neuro-fuzzy systems in construction engineering and management research, Autom. Constr. 119 (2020), https://doi.org/10.1016/j.autcon.2020.103348.
[171] J. Loy-Benitez, M.K. Song, Y.-H. Choi, J.-K. Lee, S.S. Lee, Breaking new ground: opportunities and challenges in tunnel boring machine operations with integrated management systems and artificial intelligence, Autom. Constr. 158 (2024) 105199, https://doi.org/10.1016/j.autcon.2023.105199.
[172] V.A. Cotella, From 3D point clouds to HBIM: application of artificial intelligence in cultural heritage, Autom. Constr. 152 (2023), https://doi.org/10.1016/j.autcon.2023.104936.
[173] P. Lin, M. Wu, L. Zhang, Probabilistic safety risk assessment in large-diameter tunnel construction using an interactive and explainable tree-based pipeline optimization method, Appl. Soft Comput. 143 (2023), https://doi.org/10.1016/j.asoc.2023.110376.
[174] F. Zhang, A.P.C. Chan, A. Darko, Z. Chen, D. Li, Integrated applications of building information modeling and artificial intelligence techniques in the AEC/FM industry, Autom. Constr. 139 (2022), https://doi.org/10.1016/j.autcon.2022.104289.
[175] H. Abdirad, P. Mathur, Artificial intelligence for BIM content management and delivery: case study of association rule mining for construction detailing, Adv. Eng. Inform. 50 (2021), https://doi.org/10.1016/j.aei.2021.101414.
[176] S.K. Baduge, S. Thilakarathna, J.S. Perera, M. Arashpour, P. Sharafi, B. Teodosio, A. Shringi, P. Mendis, Artificial intelligence and smart vision for building and construction 4.0: machine and deep learning methods and applications, Autom. Constr. 141 (2022) 104440, https://doi.org/10.1016/j.autcon.2022.104440.
[177] P. Kohl, O. Schmidts, L. Klöser, H. Werth, B. Kraft, A. Zündorf, STAMP 4 NLP – an agile framework for rapid quality-driven NLP applications development, Commun. Comput. Inf. Sci. (2021) 156–166, https://doi.org/10.1007/978-3-030-85347-1_12.
[178] A. Ahmad, C. Feng, M. Khan, A. Khan, A. Ullah, S. Nazir, A. Tahir, A systematic literature review on using machine learning algorithms for software requirements identification on stack overflow, Secur. Commun. Netw. 2020 (2020) 1–19, https://doi.org/10.1155/2020/8830683.
[179] H.T.Q. Nguyen, K.T.Q. Nguyen, T.C. Le, G. Zhang, Review on the use of artificial intelligence to predict fire performance of construction materials and their flame retardancy, Molecules 26 (2021) 1022, https://doi.org/10.3390/molecules26041022.
[180] E. Peterson, B. Bogosian, J. Tubella, S. Vassigh, Teaching Robotics with Virtual Reality: Developing Curriculum for the 21st Century Workforce, in: N. S., A. T.Z., K. W. (Eds.), Lect. Notes Networks Syst, Springer Science and Business Media Deutschland GmbH, Architecture Department, Florida International University, 11200 SW 8th Street, Miami, 33199, FL, United States, 2021, pp. 12–18, https://doi.org/10.1007/978-3-030-80000-0_2.
[181] M.K. Habib, S. Wagner, D. Graziotin, Detecting requirements smells with deep learning: experiences, challenges and future work, in: IEEE 29th Int. Requir. Eng. Conf. Work, 2021, IEEE, 2021, pp. 153–156, https://doi.org/10.1109/REW53955.2021.00027.
[182] B. Manzoor, I. Othman, S. Durdyev, S. Ismail, M. Wahab, Influence of artificial intelligence in civil engineering toward sustainable development—a systematic literature review, Appl. Syst. Innov. 4 (2021) 52, https://doi.org/10.3390/asi4030052.
[183] B. Manzoor, I. Othman, S. Durdyev, S. Ismail, M.H. Wahab, Influence of artificial intelligence in civil engineering toward sustainable development—a systematic literature review, Appl. Syst. Innov. (2021), https://doi.org/10.3390/asi4030052.
[184] B. Weber-Lewerenz, Corporate digital responsibility (CDR) in construction engineering—ethical guidelines for the application of digital transformation and artificial intelligence (AI) in user practice, SN Appl. Sci. 3 (2021) 801, https://doi.org/10.1007/s42452-021-04776-1.

[185] C. Debrah, A.P.C. Chan, A. Darko, Artificial intelligence in green building, Autom. Constr. 137 (2022) 104192, https://doi.org/10.1016/j.autcon.2022.104192.
[186] F. Zhang, A.P.C. Chan, A. Darko, Z. Chen, D. Li, Integrated applications of building information modeling and artificial intelligence techniques in the AEC/FM industry, Autom. Constr. 139 (2022) 104289, https://doi.org/10.1016/j.autcon.2022.104289.
[187] A.O. Onososen, I. Musonda, Perceived benefits of automation and artificial intelligence in the AEC sector: an interpretive structural modeling approach, Front. Built Environ. 8 (2022), https://doi.org/10.3389/fbuil.2022.864814.
[188] A.O. Onososen, I. Musonda, Perceived benefits of automation and artificial intelligence in the AEC sector: an interpretive structural modeling approach, Front. Built Environ. (2022), https://doi.org/10.3389/fbuil.2022.864814.
[189] N. Xu, H. Chang, B. Xiao, B. Zhang, J. Li, T. Gu, Relation extraction of domain knowledge entities for safety risk management in metro construction projects, Buildings 12 (2022) 1633, https://doi.org/10.3390/buildings12101633.
[190] M.R. Tanniru, N. Agarwal, A. Sokan, S. Hariri, An agile digital platform to support population health—a case study of a digital platform to support patients with delirium using IoT, NLP, and AI, Int. J. Environ. Res. Public Health 18 (2021) 5686, https://doi.org/10.3390/ijerph18115686.
[191] N.V. Maksimov, O.L. Golitsina, K.V. Monankov, N.A. Bal, Graph-Ontology Model of Cognitive-Similar Information Retrieval (on the Requirements Tracing Task Example), in: S. A.V., G. R.R., S. A.d. (Eds.), Adv. Intell. Syst. Comput., Springer Science and Business Media Deutschland GmbH, National Research Nuclear University MEPhI, Moscow, Russian Federation, 2021: pp. 218–224. ⟨https://doi.org/10.1007/978-3-030-65596-9_28⟩.
[192] A. Baghbani, T. Choudhury, S. Costa, J. Reiner, Application of artificial intelligence in geotechnical engineering: a state-of-the-art review, EarthSci. Rev. 228 (2022) 103991, https://doi.org/10.1016/j.earscirev.2022.103991.
[193] M. Locatelli, L.C. Tagliabue, G.M. Di Giuda, ArchiBERTo: a hierarchization quality objectives NLP Tool in the Italian architecture, engineering and construction sector, CEUR Workshop Proc. (2022).
[194] L. BAŞARIR, Modelling AI in Architectural Education, Gazi Univ. J. Sci. 35 (2022) 1260–1278, https://doi.org/10.35378/gujs.967981.
[195] M. Geva, A. Caciularu, K. Wang, Y. Goldberg, Transformer Feed-Forward Layers Build Predictions by Promoting Concepts in the Vocabulary Space. Proc. 2022 Conf. Empir. Methods Nat. Lang. Process, Association for Computational Linguistics, Stroudsburg, PA, USA, 2022, pp. 30–45, https://doi.org/10.18653/v1/2022.emnlp-main.3.
[196] Z. Zheng, X.-Z. Lu, K.-Y. Chen, Y.-C. Zhou, J.-R. Lin, Pretrained domain-specific language model for natural language processing tasks in the AEC domain, Comput. Ind. 142 (2022) 103733, https://doi.org/10.1016/j.compind.2022.103733.
[197] D. Mackenzie, Surprising advances in generative artificial intelligence prompt amazement—and worries, Engineering 25 (2023) 9–11, https://doi.org/10.1016/j.eng.2023.04.004.
[198] L. Blum, M. Blum, A theoretical computer science perspective on consciousness and artificial general intelligence, Engineering 25 (2023) 12–16, https://doi.org/10.1016/j.eng.2023.03.010.
[199] X. Qu, Y. Gu, Q. Xia, Z. Li, Z. Wang, B. Huai, A survey on arabic named entity recognition: past, recent advances, and future trends, IEEE Trans. Knowl. Data Eng. (2023) 1–18, https://doi.org/10.1109/TKDE.2023.3303136.
[200] P.V. Thayyib, R. Mamilla, M. Khan, H. Fatima, M. Asim, I. Anwar, M. K. Shamsudheen, M.A. Khan, State-of-the-art of artificial intelligence and big data analytics reviews in five different domains: a bibliometric summary, Sustainability 15 (2023) 4026, https://doi.org/10.3390/su15054026.
[201] Y. Pan, L. Zhang, Integrating BIM and AI for smart construction management: current status and future directions, Arch. Comput. Methods Eng. 30 (2023) 1081–1110, https://doi.org/10.1007/s11831-022-09830-8.
[202] H. Afzal, P. Buitelaar, T. Wunner, P. Cimiano, J. McCrae, Generating lexical information for terminology in a bioinformatics ontology, in: TKE 2010 Present. Terminol. Knowl. Eng. Resour. Online Model. Challenges, Department of Computer Science, College of Telecommunication Engineering, National University of Sciences and Technology, Pakistan, 2010: pp. 56–76.
[203] L. Kof, From Requirements Documents to System Models: A Tool for Interactive Semi-Automatic Translation, in: 2010 18th IEEE Int. Requir. Eng. Conf., IEEE, Fakultaet fuer Informatik, Technische Universitaet Muenchen, D-85748 Garching bei Muenchen, Boltzmannstr. 3, Germany, 2010: pp. 391–392. ⟨https://doi.org/10.1109/RE.2010.53⟩.
[204] B. Wu, Y. Song, J. Chu, J. Cao, Information Extraction for Knowledge Graph of ISO 19650 Standards, in: J. L.C., K. R., Z. K., K. R. (Eds.), Smart Innov. Syst. Technol, Springer Science and Business Media Deutschland GmbH, Economic Research Institute of State Grid Zhejiang Electrical Power Company, Hangzhou, 2022, pp. 162–172, https://doi.org/10.1007/978-981-19-3486-5_20.
[205] S. Shooshtarian, A.T. Gurmu, A.-M. Sadick, Application of natural language processing in residential building defects analysis: Australian stakeholders' perceptions, causes and types, Eng. Appl. Artif. Intell. 126 (2023) 107178, https://doi.org/10.1016/j.engappai.2023.107178.
[206] H. Nabizadeh Rafsanjani, A.H. Nabizadeh, Towards human-centered artificial intelligence (AI) in architecture, engineering, and construction (AEC) industry, Comput. Hum. Behav. Rep. 11 (2023) 100319, https://doi.org/10.1016/j.chbr.2023.100319.
[207] S. Moon, G. Lee, S. Chi, Corrigendum to "Automated system for construction specification review using natural language processing" [Adv. Eng. Inf. 51 (2022) 101495], Adv. Eng. Inform. 52 (2022) 101626, https://doi.org/10.1016/j.aei.2022.101626.
[208] Y. Wang, J. Chen, X. Xia, B. Jiang, Intelligent requirements elicitation and modeling: a literature review, Jisuanji Yanjiu yu Fazhan, Comput. Res. Dev. 58 (2021) 683–705, https://doi.org/10.7544/issn1000-1239.2021.20200740.
[209] N. Wang, R.R.A. Issa, Natural language generation from building information models for intelligent NLP-based information extraction, in: EG-ICE, 2020 Work. Intell. Comput. Eng. Proc. (2020).
[210] J. Wang, X. Gao, X. Zhou, Q. Xie, Multi-scale Information Retrieval for BIM using Hierarchical Structure Modelling and Natural Language Processing, J. Inf. Technol. Constr. 26 (2021) 409–426, https://doi.org/10.36680/j.itcon.2021.022.
[211] M. Saarela, V. Podgorelec, Recent applications of explainable AI (XAI): a systematic literature review, Appl. Sci. 14 (2024), https://doi.org/10.3390/app14198884.
[212] N. Musa, A. Ibrahim, M. Tiwari, Explainable Artificial Intelligence (XAI) Techniques To Enhance Transparency In Deep Learning Models, (2024) 29–36. ⟨https://doi.org/10.9790/0661-2606012936⟩.
[213] D.M. Salama, N.M. El-Gohary, Natural language processing for automated regulatory and contractual document analysis, in: Proceedings, Annu. Conf. - Can. Soc. Civ. Eng., Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, Urbana, IL, United States, 2011: pp. 2897–2906.
[214] D. Guo, E. Onstein, A.D.La Rosa, A semantic approach for automated rule compliance checking in construction industry, IEEE Access 9 (2021) 129648–129660, https://doi.org/10.1109/ACCESS.2021.3108226.
[215] M.A.A.S. Ali, AI-natural language processing (NLP), Int. J. Res. Appl. Sci. Eng. Technol. 9 (2021) 135–140, https://doi.org/10.22214/ijraset.2021.37293.
[216] S. Mokarizadeh, P. Küngas, M. Matskin, Ontology learning for cost-effective large-scale semantic annotation of web service interfaces. Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), Royal Institute of Technology (KTH), Stockholm, Sweden, 2010, pp. 401–410, https://doi.org/10.1007/978-3-642-16438-5_30.
[217] Y. Jiang, H. Dong, H. Xiong, OntoLTCn: A Chinese Text Oriented Semi-auto Ontology Knowledge Discovery Tool, in: 2008 Int. Conf. Comput. Sci. Softw. Eng., IEEE, School of Information Management, Wuhan University, Wuhan, 420072, China, 2008: pp. 662–665. ⟨https://doi.org/10.1109/CSSE.2008.788⟩.
[218] H. Xu, Coordinated distribution of stadium electric load unloading based on artificial intelligence, Energy Rep. 8 (2022) 10594–10603, https://doi.org/10.1016/j.egyr.2022.08.207.
[219] C. Islam, M.A. Babar, R. Croft, H. Janicke, SmartValidator: a framework for automatic identification and classification of cyber threat data, J. Netw. Comput. Appl. 202 (2022) 103370, https://doi.org/10.1016/j.jnca.2022.103370.
[220] A.F. Romero Gomez, A.D. Orjuela-Canon, A.L. Jutinico, C. Awad, E. Vergara, A. Palencia, Preliminary Text Analysis from Medical Records for TB Diagnosis Support, in: 2021 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc., IEEE, Universidad Del Rosario, Biomedical Engineering Program, Escuela Colombiana de Ingeniería Julio Garavito, Bogotá, Colombia, 2021: pp. 2468–2471. ⟨https://doi.org/10.1109/EMBC46164.2021.9631006⟩.
[221] D.M. Salama, N.M. El-Gohary, Semantic Modeling for Automated Compliance Checking, in: Comput. Civ. Eng., American Society of Civil Engineers, Reston, VA, University of Illinois at Urbana-Champaign, Urbana, IL 61801, 205 North Mathews Ave., United States, 2011: pp. 641–648. ⟨https://doi.org/10.1061/41182(416)79⟩.
[222] R. Kessler, N. Béchet, Deep Learn. Nat. Lang. Process. Field Constr. (2025). ⟨https://arxiv.org/abs/2501.07911v1⟩.
[223] Yao Liu, Zhifang Sui, Yongwei Hu, Qingliang Zhao, Research on Automatic Construction of Medical Ontology, in: 2010 Int. Conf. Biomed. Eng. Comput. Sci., IEEE, Institute of Scientific and Technical Information of China, Beijing, China, 2010, pp. 1–4, https://doi.org/10.1109/ICBECS.2010.5462404.
[224] J. Wang, L. Li, L. Long, Research of the method for obtaining semantic knowledge in the system of CI-NLU based text processing after speech recognition, in: 2009 Int. Conf. Nat. Lang. Process. Knowl. Eng., IEEE, Beijing University of Posts and Telecommunications, Beijing, China, 2009, pp. 1–6, https://doi.org/10.1109/NLPKE.2009.5313779.
[225] S. Nicoll, K. Douglas, C. Brinton, Giving feedback on feedback: an assessment of grader feedback construction on student performance. LAK22 12th Int. Learn. Anal. Knowl. Conf, ACM, New York, NY, USA, 2022, pp. 239–249, https://doi.org/10.1145/3506860.3506897.
[226] W.A.M.B. Ahmad Tarmizi, A.N. Zanol Rashid, A.A. Muhammad SAPRI, M. Yangkatisal, Natural language processing (NLP) application For classifying and managing tacit knowledge in revolutionizing ai-driven library, Inf. Manag. Bus. Rev. 16 (2024) 1094–1110, https://doi.org/10.22610/imbr.v16i3(i)s.3949.
[227] Y. Kano, M. Miwa, K.B. Cohen, L.E. Hunter, S. Ananiadou, J. Tsujii, U-Compare: a modular NLP workflow construction and evaluation system, IBM J. Res. Dev. 55 (11) (2011) 1–11, https://doi.org/10.1147/JRD.2011.2105691.
[228] M. Salim Keezhatta, Understanding EFL linguistic models through relationship between natural language processing and artificial intelligence applications, Arab World Engl. J. 10 (2019) 251–262, https://doi.org/10.24093/awej/vol10no4.19.